

Computational Psychodynamic Theory

Ajith Senthil

December 2024

Abstract

This paper introduces Computational Psychodynamics, a novel mathematical framework for modeling cognition, personality, and behavior. Drawing inspiration from computational linguistics' use of statistical models for linguistic analysis, our approach employs a hierarchical behavioral schema based on linearly separable binary classifications. The framework integrates the Free Energy Principle and Active Inference (FEP-ActInf) with established psychological theories, providing a mathematical foundation for understanding behavioral dynamics through the principle of least action. By modeling transitions between behavioral states using time-inhomogeneous Markov chains and hidden Markov models, Computational Psychodynamics enables sophisticated analysis of both observable behaviors and latent psychological states. The framework introduces observation-based psychometrics grounded in ecological psychology, offering new approaches to personality assessment and behavioral prediction. I demonstrate how transfer entropy between behavioral chains can reveal patterns of attachment and social influence while providing insights into personality dynamics aligned with established theories such as the Big Five. Applications of this framework range from generating realistic character behaviors in narrative environments to developing brain-computer interfaces and analyzing social network dynamics. Through its integration of psychological theory with rigorous mathematical modeling, Computational Psychodynamics offers a comprehensive approach to understanding and predicting human behavior, with implications for both theoretical research and practical applications in psychology, artificial intelligence, and related fields.

1 Introduction

The field of psychology has long sought to develop comprehensive frameworks for understanding human behavior, cognition, and personality. Traditional approaches have typically relied on self-report measures, subjective observations, and compartmentalized theories that address specific aspects of human psychology in isolation. While these methods have provided valuable insights, they often fall short of capturing the dynamic, interconnected nature of human behavior and cognition in real-world contexts.

Current psychological frameworks face several significant challenges. First, there is a notable disconnect between theoretical models and their practical implementation, particularly in computational contexts. Second, existing approaches often lack the mathematical rigor necessary for precise modeling and prediction of behavioral dynamics. Third, traditional psychometric tools, while useful, often fail to capture the ecological validity of personality and behavior as they manifest in natural environments (Satchell, Kaaronen, & Latzman, 2020).

The emergence of ecological psychology has highlighted the importance of understanding behavior within its environmental context, emphasizing how organisms actively make sense of their surroundings through the perception of affordances (Gibson, 1979). Simultaneously, cybernetic theories have provided insights into goal-directed behavior and self-regulation processes. However, these approaches have largely remained separate from mainstream psychological assessment and modeling techniques.

Recent advances in Active Inference theory (Parr, Friston, et al., 2020) have provided a promising framework for understanding cognition as a process of free energy minimization, offering a mathematical foundation for modeling how agents interact with their environment. This development, combined with the growing need for more sophisticated approaches to personality assessment and behavioral prediction, calls for a new integrated framework.

Computational Psychodynamics is a novel mathematical framework that synthesizes insights from computational linguistics, ecological psychology, and Active Inference theory in response to these challenges. Drawing inspiration from computational linguistics and applying statistical models to derive linguistic metrics, our approach introduces a hierarchical behavioral schema based on linearly separable binary classifications. This schema provides a structured foundation for understanding and modeling behavior, similar to how parts of speech serve as fundamental categories in linguistics.

Computational Psychodynamics offers several key innovations:

1. Integration of ecological and cybernetic perspectives through a mathematical framework that captures the dynamic interaction between agents and their environment
2. Development of observation-based psychometrics that leverages natural behavioral data rather than relying solely on self-report measures
3. Implementation of sophisticated statistical tools, including Markov chains and Transformer models, for modeling behavioral state transitions
4. Application of Active Inference principles to understand and predict personality expression and behavioral dynamics

This paper outlines the theoretical foundations of Computational Psychodynamics, outlines its practical applications in areas ranging from artificial intelligence to clinical psychology, and provides a roadmap for future empirical research and implementation. Computational Psychodynamics offers a promising approach to advancing our understanding of human behavior and cognition by bridging the gap between theoretical frameworks and computational implementation.

1.1 Observation-Based Psychometrics: An Ecological Application of Computational Psychodynamics

One promising application of Computational Psychodynamics lies in the development of observation-based psychometrics. Traditional personality assessment methods often rely heavily on self-report measures, which can be subject to various biases and limitations. The ecological approach to personality assessment, as advocated by Satchell, Kaaronen, and Latzman (2020), offers a more naturalistic and objective framework for understanding personality through behavioral observation and environmental interaction.

1.1.1 The Ecological Perspective on Personality Assessment

The foundation of observation-based psychometrics stems from the ecological psychology tradition, which emphasizes the importance of studying behavior in context rather than in isolation. In this view, personality traits are not merely internal characteristics but manifest as patterns of interaction between individuals and their environment. Satchell et al. (2020) propose that personality traits can be understood as behavioral methods for processing environmental information in relation to oneself. This perspective aligns with Gibson’s (1979) concept of affordances, where the environment presents action possibilities that are perceived differently based on individual characteristics.

1.1.2 Benefits of an Ecological Approach

The ecological approach to personality assessment offers several advantages over traditional methods:

1. **Cross-Species Applicability:** By focusing on observable behaviors and environmental interactions, this approach enables both interspecies and intraspecies personality modeling, providing a more comprehensive framework for understanding personality across different contexts.
2. **Reduced Measurement Bias:** Observation-based methods can help minimize the self-report biases inherent in traditional personality assessments, as they rely on direct behavioral observation rather than subjective self-evaluation.

3. Context Sensitivity: The approach naturally accounts for the role of environmental context in personality expression, acknowledging that behavior emerges from the interaction between individual traits and situational factors.

1.1.3 Implementation Through Computational Psychodynamics

Computational Psychodynamics provides the mathematical framework necessary to implement observation-based psychometrics effectively. By leveraging Active Inference principles (Parr, Friston, et al., 2020), the framework can model how individuals actively make sense of their environment and how their personality traits influence this process. This computational approach allows for:

- Quantitative modeling of the relationship between personality traits and behavioral patterns
- Integration of multiple data sources, including environmental variables and observed behaviors
- Dynamic updating of personality assessments based on ongoing behavioral observations

1.1.4 Future Directions

The development of observation-based psychometrics through Computational Psychodynamics addresses the call by Möttus et al. (2020) for innovative approaches in psychometric tool development. This integration of ecological psychology, Active Inference, and computational modeling offers promising avenues for advancing our understanding of personality and improving assessment methods. Future research should focus on validating these observation-based measures against traditional assessments while exploring their potential applications in clinical settings, organizational behavior, and artificial intelligence development.

2 Simulation and Personality Modeling Through Computational Psychodynamics

One application of Computational Psychodynamics lies in its capacity to both simulate personality dynamics and model existing personalities through observational data. This dual capability enables both the generation of realistic character behaviors and the capture of real personality patterns, offering significant implications for fields ranging from narrative generation to psychological assessment.

2.1 Personality Simulation Framework

Here I will present one of the many possible simulation frameworks within Computational Psychodynamics that utilizes a combination of Markov transition matrices and the hierarchical behavioral schema that will be introduced. This specific approach enables:

1. State-Based Behavioral Modeling:
 - Implementation of transition matrices that govern probabilistic shifts between behavioral states
 - Integration of hierarchical behavioral classes that mirror natural personality expression
 - Dynamic updating of behavioral probabilities based on context and previous states
2. Realistic Character Generation:
 - Creation of consistent personality patterns through stationary distribution analysis
 - Generation of contextually appropriate responses using prototype behavioral classes
 - Maintenance of personality coherence across different situations and interactions

2.2 Personality Capture and Modeling

The framework also provides mechanisms for modeling existing personalities through observational data:

1. Data Collection and Processing:
 - Systematic collection of behavioral observations over time
 - Integration of multiple data sources, including textual descriptions and video recordings
 - Application of AI technologies for automated behavioral annotation
2. Model Construction:
 - Development of individualized transition matrices from observed behavioral patterns
 - Implementation of Hidden Markov Models (HMMs) to capture underlying affective states
 - Creation of prototype classes corresponding to observed behavioral states

2.3 Analytical Capabilities

The framework offers several analytical tools for understanding and validating personality models:

1. Dynamic Analysis:
 - Evaluation of behavioral sequence patterns
 - Assessment of transition probabilities between states
 - Calculation of entropy rates to measure behavioral predictability
2. Temporal Patterns:
 - Analysis of mean first passage times between states
 - Identification of transient versus recurrent behavioral patterns
 - Prediction of likely behavioral progressions

2.4 Applications and Implementation

This dual simulation-modeling capability finds practical applications in various domains:

1. Narrative Generation:
 - Creation of believable character behaviors in interactive storytelling
 - Generation of consistent personality-driven narratives
 - Development of dynamic character interactions
2. Psychological Assessment:
 - Creation of baseline personality models for comparative analysis
 - Development of personalized intervention strategies
 - Tracking of behavioral changes over time
3. Bi-Directional Brain Computer Interfacing:
 - Use reinforcement learning to read and write data to minds through robotic controlled Transcranial Magnetic Stimulation (TMS)
 - Use reinforcement learning to predict optimal behavioral interventions or treatments
4. AI and Robotics:
 - Implementation of personality-driven decision-making in artificial agents
 - Development of more naturalistic human-robot interactions
 - Creation of adaptive behavioral models

2.5 Future Directions

The simulation and modeling capabilities of Computational Psychodynamics open several avenues for future research and development:

1. Integration with Advanced AI:

- Enhancement of behavioral observation through computer vision
- Implementation of more sophisticated event understanding models
- Development of real-time personality modeling systems

2. Clinical Applications:

- Development of personalized treatment models
- Creation of predictive behavioral tools
- Develop digital pharmaceuticals using artificial intelligence
- Implementation of adaptive intervention strategies
- Develop high precision TMS treatments and bi-directional computational interfacing with the mind

3. Virtual/Augmented Reality and Interfacing with Dreams:

- Interface computationally with the mind to develop methods of interfacing with dreams and sharing dreams between multiple people
- Interface computationally with the mind to develop methods of enhancing emotional connection and learning
- Interface computationally with the mind to develop virtual/augmented reality experiences

This framework provides a foundation for both generating realistic personality-driven behaviors and capturing existing personality patterns through observation, offering tools for both theoretical research and practical applications in psychology, artificial intelligence, and related fields.

3 Theoretical Framework Connections in Binary Cognitive Mechanics

3.1 Binary Cognitive Mechanics and Psychological Frameworks

3.1.1 Foundational Integration with Jung's Cognitive Functions

Binary Cognitive Mechanics (BCM) reinterprets Jung's Cognitive Functions in an information-theoretic and combinatorial framework (?). Mathematically, it relies on an $S(3, 4, 8)$ Steiner system, which is defined on a *universal set* U of eight elements. Each *block* in this system is a 4-element subset $B_i \subset U$. By design:

- Every 3-element subset of U occurs in exactly one block.
- There are exactly 14 such blocks in total.

A notable property of $S(3, 4, 8)$ is that each block B_i has a *complementary block* B_j such that

$$B_i \cup B_j = U \quad \text{and} \quad B_i \cap B_j = \emptyset.$$

BCM identifies seven such pairs, each pair corresponding to one of the seven key *dichotomies* (e.g., Introverted vs. Extraverted). These seven dichotomies, in turn, combine to generate the eight cognitive functions originally described by Jung.

3.1.2 Understanding Entropy in BCM

Before discussing how BCM links to other psychological frameworks, we first clarify how *entropy* is defined and used within BCM. In *information theory*, introduced by Shannon, the entropy of a probability distribution $p(x)$ over discrete states $\{x_1, x_2, \dots\}$ is given by:

$$H[p] = - \sum_i p(x_i) \log[p(x_i)].$$

In **BCM**, we interpret x_i as a possible *representational state* of a cognitive system (e.g., ways of encoding sensory data, forming beliefs, or constructing plans). The probability $p(x_i)$ reflects how likely the system is to adopt or maintain state x_i .

High- vs. Low-Entropy Cognition. When the distribution $p(x)$ is *broad*—assigning relatively uniform probabilities to many states—the entropy is *high*. BCM associates this scenario with “extraverted” or “variety-seeking” orientations, in which the agent keeps multiple possibilities or interpretations open. Conversely, when $p(x)$ is *narrow*—assigning most of its probability mass to fewer states—the entropy is *low*. BCM associates this with “introverted” or “certainty-seeking” orientations, in which the agent refines and relies on fewer, more assured representations.

Thus, while the Shannon formula itself remains the same in both cases, the *cognitive interpretation* differs according to whether the system is focusing on a narrow set of representations (low entropy) or distributing attention across many possibilities (high entropy). This idea underpins BCM’s approach to bridging various psychological theories.

3.1.3 Schwartz’s Theory of Basic Human Values

Although Schwartz’s Theory of Basic Human Values (?) is not itself information-theoretic, BCM draws parallels between:

- **Introverted attention**, interpreted as a preference for *low-entropy* (more certain) states,
- **Extraverted attention**, interpreted as a preference for *high-entropy* (more varied) states,
- **Internalizing vs. Externalizing** representations, interpreted as a distinction between *personal* vs. *social* focus.

These two dichotomies (introversion/extraversion and internalizing/externalizing) can be viewed through an entropy lens. Let $\{x_i\}$ be the system’s possible internal representations and actions, and let $p(x_i)$ quantify how the system *distributes* its representational “attention.” Then,

$$H_{\text{low}} = - \sum_i p(x_i) \log[p(x_i)] \quad \left| \begin{array}{l} \text{associated with} \\ \text{introverted (low-entropy)} \end{array} \right. , \quad (1)$$

$$H_{\text{high}} = - \sum_i p(x_i) \log[p(x_i)] \quad \left| \begin{array}{l} \text{associated with} \\ \text{extraverted (high-entropy)} \end{array} \right. . \quad (2)$$

Although Eqs. (1)–(2) share the same Shannon formula, H_{low} is conceptually tied to *narrow, well-established* cognitive states and H_{high} to *broad, flexible* states. BCM proposes that the difference

$$\Delta H_{\text{system}} = H_{\text{high}} - H_{\text{low}}$$

can map onto how an agent balances *openness* (similar to “gain-promotion” or “exploration”) against *conservation* (similar to “loss-prevention” or “certainty”)—a theme that aligns with Schwartz’s main motivational axes.

3.1.4 Cybernetic Big Five Theory and Cybernetic Value Fulfillment Theory

DeYoung’s Cybernetic Big Five Theory (CB5T) (??) interprets personality traits as parameters in an evolved *cybernetic system*, emphasizing two higher-order meta-traits: *Stability* and *Plasticity*. Although DeYoung does not define these traits via entropy, BCM suggests they can be modeled as functions of whether a system operates in a *low-entropy* regime (favoring Stability) or a *high-entropy* regime (favoring Plasticity).

Formally, let σ_s (Stability) and σ_p (Plasticity) be:

$$\begin{aligned}\sigma_s &= f(H_{\text{low}}), \\ \sigma_p &= f(H_{\text{high}}),\end{aligned}\tag{3}$$

where H_{low} and H_{high} again refer to the *distribution* of cognitive states in narrower vs. broader configurations, and f is a mapping from these entropy measures to trait-level expressions.

BCM thereby frames σ_s as reflecting how strongly an agent seeks order or certainty, and σ_p as reflecting how strongly it seeks novelty or variation. The interplay of σ_s and σ_p can then yield a set of *goal states* $\{g_i\}$:

$$G = \left\{ g_i \mid g_i = f(\sigma_s, \sigma_p) \right\},\tag{4}$$

illustrating how personality meta-traits might govern cognitive or behavioral goals in a probabilistic (i.e. entropy-based) way.

3.1.5 Self-Determination Theory Alignment

Self-Determination Theory (SDT) (?) highlights three fundamental psychological needs: *autonomy*, *competence*, and *relatedness*. While SDT does not analyze these needs in terms of probabilistic states or entropy, BCM proposes a meaningful conceptual bridge:

- **Competence** can be seen as the agent’s effectiveness at managing its representational distributions, i.e. reducing *surprise* or making accurate predictions in different entropy regimes.
- **Autonomy** aligns with *introverted decision-making* (low-entropy weighting), where the agent prioritizes internally consistent representations.
- **Relatedness** aligns with *extraverted decision-making* (high-entropy weighting), involving broader social or collective representations.

A simple mathematical illustration might let O_i and O_e be *observation functions* for introverted/extraverted attention, each tied to a particular entropy regime:

$$O_i \approx \Phi(H_{\text{low}}), \quad O_e \approx \Psi(H_{\text{high}}),$$

where Φ and Ψ describe how effectively the system can gather or refine information under narrower vs. broader states. Decision functions D_i (autonomy) and D_e (relatedness) may each depend on these observation functions, reflecting how personal or social motivations dominate at varying levels of entropy.

Overall, these examples illustrate BCM’s central claim: *cognitive processes can be represented by probability distributions over internal states, whose Shannon entropy characterizes the system’s level of certainty or flexibility*. By combining this idea with Steiner-system partitions, BCM unifies the computational basis (entropy measures and combinatorial dichotomies) with established psychological theories—Jung’s typology, Schwartz’s values, CB5T, and SDT—thereby highlighting recurring themes of *opposition and integration* within cognitive structures.

3.2 Active Inference and Cybernetic Theories Integration

3.2.1 Free Energy Principle Integration

The Free Energy Principle-Active Inference (FEP-ActInf) framework provides a computational foundation for understanding how BCM’s cognitive functions operate. Safron & DeYoung (2021) established connections between CB5T and FEP-ActInf, which Computational Psychodynamics extends through its focus on computational modeling.

This integration is formalized through the free energy equation:

$$F = D[Q(s)||P(s|a, G)] + H[Q(s)] \quad (5)$$

Where:

- F represents free energy
- $D[\bullet||\bullet]$ denotes Kullback-Leibler divergence
- $Q(s)$ represents belief states
- $P(s|a, G)$ denotes posterior probability
- $H[\bullet]$ represents entropy

3.2.2 Markov Blankets and Cognitive Functions

BCM’s interpretation of cognitive functions aligns with the Markov blanket concept in Active Inference (Kirchhoff et al., 2018). The framework demonstrates how:

- Internal states (represented by introverted functions) and external states (represented by extraverted functions) are separated by statistical boundaries
- Information processing occurs through hierarchical layers of Markov blankets
- Self-organization emerges through the minimization of free energy

3.2.3 Entropy Management in Cognitive Processing

BCM’s treatment of cognitive entropy directly parallels Active Inference’s free energy minimization principle:

- Low-entropy processes (certainty-seeking) align with introverted functions
- High-entropy processes (adaptability-seeking) correspond to extraverted functions
- The balance between these processes reflects the system’s attempt to minimize free energy while maintaining adaptive flexibility

This is formally represented through cognitive vectors:

$$\begin{aligned} O_i + D_i \text{ (Sleep)} &: \arg \min_a D[Q(s)||P(s|a, G)] + H[Q(s)] \\ O_i + D_e \text{ (Blast)} &: \arg \min_a D[Q'(s')||P(s'|a, G')] + H[Q'(s')] \\ O_e + D_i \text{ (Consume)} &: \arg \max_a D[Q(s)||P(s|a, G)] + H[Q(s)] \\ O_e + D_e \text{ (Play)} &: \arg \max_a D[Q'(s')||P(s'|a, G')] + H[Q'(s')] \end{aligned} \quad (6)$$

3.2.4 Hierarchical Cognitive Processing

The framework integrates hierarchical processing across multiple levels:

- Individual cognitive functions
- Function pairs (“Animals” or action types)
- Overall personality system architecture

This hierarchical structure aligns with Active Inference’s treatment of nested generative models and prediction error minimization across different temporal and spatial scales.

3.3 Synthesis and Implications

3.3.1 Unified Framework for Personality and Cognition

BCM provides a unified framework that integrates:

- Information-theoretic principles
- Cybernetic control mechanisms
- Active inference processes
- Traditional personality theories

3.3.2 Practical Applications

The integration of these frameworks enables:

- More precise personality assessment methods
- Better understanding of cognitive development
- Improved modeling of decision-making processes through state space definition:

$$S = \{s_i | s_i = f(CV_i, H_i, F_i)\} \quad (7)$$

Where s_i represents system states, CV_i cognitive vectors, H_i entropy states, and F_i free energy states

- Enhanced prediction of behavioral patterns

3.3.3 Future Research Directions

Key areas for future investigation include:

- Empirical validation of BCM’s predictions about cognitive function interactions
- Development of computational models incorporating BCM’s principles
- Investigation of temporal dynamics in personality expression
- Application to artificial intelligence and cognitive architecture design

This synthesis of theories provides a robust foundation for understanding human cognition and personality through the lens of information processing and cybernetic control, while maintaining compatibility with established psychological frameworks.

4 Steiner Systems and Their Relevance to Behavioral Modeling

4.1 Introduction and Motivation

Modeling complex behavior requires a framework that can uniquely classify states while minimizing redundancy and ambiguity. In many domains—from cognitive science to language analysis—the challenge is to capture the nuanced differences between behaviors with a robust, mathematically grounded structure. Our approach leverages the remarkable properties of Steiner systems, particularly the $S(3, 4, 8)$ design, to meet these challenges.

Steiner systems are finite combinatorial designs characterized by the unique coverage of every t -element subset within a collection of k -element blocks. In the case of $S(3, 4, 8)$, each triple of points is contained in exactly one of 14 blocks. This property guarantees:

- **Mutual Exclusivity:** Each behavioral state (or cognitive function) is uniquely represented, ensuring that no two states ambiguously share the same defining features.
- **Finite and Exhaustive Coverage:** Every possible combination of key components is represented exactly once, providing a complete and balanced classification system.
- **Rich Symmetry:** The inherent automorphism group of the Steiner system supports natural reorientations and pairings of states (e.g., defining opposites or complementary pairs) without loss of structure.

These combinatorial features form the backbone of our Binary Cognitive Mechanics (BCM) framework, which maps the eight points of $S(3, 4, 8)$ onto Jung’s cognitive functions and extracts seven fundamental dichotomies through strategic block pairing.

Embedding this combinatorial structure into a geometric framework—using ideas from hypersphere packing and vector space representations—further enhances our model. By associating each behavioral category with a hypersphere in an optimally embedded space, we can:

- Translate discrete combinatorial distinctions into continuous spatial relationships.
- Exploit linear transformations to capture partial information, where “not-opposite” constraints help narrow down classifications even under uncertainty.
- Define state transitions via Markov chains, with transition probabilities informed by the geometric and combinatorial overlap of the blocks.

In essence, this unified framework not only provides optimal behavioral space partitioning but also lays a theoretical foundation for efficient communication. Minimal linguistic signals—capable of excluding one block or its complement—are shown to be sufficient for influencing behavioral trajectories, a result with far-reaching implications for understanding how information is encoded and transferred in cognitive and social systems.

The following sections will detail the mathematical properties of Steiner systems, the geometric embedding techniques, and the stochastic and communicative models that arise from this integrated approach.

4.2 Combinatorial Foundations: Steiner Systems

This section establishes the combinatorial groundwork by defining Steiner systems and highlighting their fundamental properties. In particular, we focus on the Steiner system $S(t, k, v)$ and its special instance $S(3, 4, 8)$, which plays a central role in our modeling framework.

4.2.1 Definition and Fundamental Properties

A **Steiner system** is a finite combinatorial design specified by three integers (t, k, v) with $t < k < v$. It is formally defined as follows:

1. **Set and Blocks:** Let X be a finite set with $|X| = v$ elements (often referred to as “points”). Let $\mathcal{B}$ be a collection of subsets of X (called “blocks”), each having exactly k elements.

2. **Unique Coverage Property:** Every t -element subset $Y \subseteq X$ appears in exactly one block $B \in \mathcal{B}$. This can be expressed mathematically as:

$$S(t, k, v) \iff \begin{cases} |X| = v, \\ B \in \mathcal{B}, \quad |B| = k, \\ \forall Y \subseteq X, |Y| = t \implies \exists! B \in \mathcal{B} : Y \subseteq B. \end{cases}$$

The requirement that every t -subset appears in exactly one block imposes a strong uniqueness constraint, which leads to:

- **Unique Triple Coverage:** For $t = 3$, every triple of points is contained in one and only one block.
- **Minimal Overlap:** No two blocks share the same t -subset; that is, any two distinct blocks can intersect in at most $t - 1$ points.

4.2.2 Key Properties of $S(3, 4, 8)$

The Steiner system $S(3, 4, 8)$ is a concrete example with the following parameters:

- **$v = 8$** (number of points)
- **$k = 4$** (size of each block)
- **$t = 3$** (each triple is uniquely covered)

Using the classical formula for the number of blocks,

$$b = \frac{\binom{v}{t}}{\binom{k}{t}},$$

we compute:

$$b = \frac{\binom{8}{3}}{\binom{4}{3}} = \frac{56}{4} = 14.$$

Thus, there are exactly 14 blocks in $S(3, 4, 8)$.

Additional noteworthy properties include:

- **Rich Symmetry and Automorphism Group:** The system possesses a large automorphism group, meaning that many permutations of the 8 points map blocks to blocks without altering the overall structure. This symmetry:
 - Justifies the reorientation or relabeling of points without loss of combinatorial integrity.
 - Provides a natural foundation for pairing blocks (e.g., defining opposites or complements), which is instrumental in later modeling steps.
- **Combinatorial Balance:** The exact coverage condition ensures that every triple appears exactly once, leading to minimal ambiguity and avoiding redundant overlaps among blocks. Such balance is essential when translating combinatorial configurations into robust behavioral categories.

In summary, the Steiner system $S(3, 4, 8)$ exemplifies a finite, balanced, and highly symmetric structure where every triple of points is uniquely assigned to a block. These properties simplify the classification problem by ensuring minimal overlap and facilitate natural pairings and reconfigurations—features that are crucial for subsequent geometric embedding and stochastic modeling in our framework.

5 Active Inference Integration in Computational Psychodynamics

5.1 Fundamental Behavioral Categories in Active Inference

The framework of Computational Psychodynamics (CompPsycho) extends Binary Cognitive Mechanics through Active Inference by formalizing cognitive functions as specific optimization objectives in the free-energy landscape. Each cognitive vector pair (“Animal”) corresponds to distinct free-energy minimization or maximization strategies.

5.1.1 Notation and Setup

Generative Models. Let M^i denote the generative model maintained by agent i . This model specifies how observations o^i (and possibly states s^i) depend on actions a^i and relevant parameters. Agent i forms a posterior $Q^i(s^i)$ over states, based on o^i .

Thus, for two agents ($i = 1, 2$) in a dyadic setting, we might have (s^1, o^1, a^1, M^1) for agent 1 and (s^2, o^2, a^2, M^2) for agent 2, but this extends naturally to any number of agents i .

5.1.2 Mathematical Formalization of “Animals”

For an agent i interacting with its environment, define:

s^i : External (hidden) states of the world (for agent i), o^i : Observations perceived by agent i , a^i : Agent i ’s action space

$Q^i(s^i)$: Agent i ’s (approximate) posterior beliefs about states, $P^i(s^i | a^i, M^i)$: Model-based posterior or likelihood under

Remark: In some Active Inference expositions, $P^i(s^i | o^i, a^i, M^i)$ might be standard. For simplicity, we use $P^i(s^i | a^i, M^i)$ to emphasize the agent’s update under different actions. If we had a strictly Bayesian formulation, we’d have $P^i(s^i | o^i, a^i, M^i)$.

If we consider an *other* agent j , we analogously have $(s^j, o^j, a^j, M^j, Q^j(s^j))$, etc.

Dyadic Example. For clarity, if we only want to model *two* agents (a truly dyadic setting), we might restrict $i = 1$ and $j = 2$. Then agent 1 can attempt to minimize or maximize agent 2’s free-energy, etc. If we want a general multi-agent scenario, the same notation extends to $i = 1, 2, 3, \dots$

Four Fundamental Behavioral Categories (“Animals”) Now, define four modes of free-energy–based objectives:

1. Sleep/Revising ($O_i + D_i$)

$$a^i = \arg \min_{a^i} \left[D(Q^i(s^i) \| P^i(s^i | a^i, M^i)) + H(Q^i(s^i)) \right].$$

Represents *self-directed* uncertainty reduction for agent i .

2. Blast/Directing ($O_i + D_e$)

$$a^i = \arg \min_{a^i} \left[D(Q^j(s^j) \| P^j(s^j | a^i, M^j)) + H(Q^j(s^j)) \right].$$

Represents *other-directed* uncertainty reduction, i.e. agent i chooses a^i to reduce agent j ’s free energy or mismatch.

3. Consume/Seeking ($O_e + D_i$)

$$a^i = \arg \max_{a^i} \left[D(Q^i(s^i) \| P^i(s^i | a^i, M^i)) + H(Q^i(s^i)) \right].$$

Represents *self-directed* uncertainty *increase* for agent i .

4. Play/Conferring ($O_e + D_e$)

$$a^i = \arg \max_{a^i} \left[D(Q^j(s^j) \parallel P^j(s^j \mid a^i, M^j)) + H(Q^j(s^j)) \right].$$

Represents *other-directed* uncertainty *increase* (shake things up for agent j).

5.2 Information-Processing Axes

5.2.1 N-S Axis Formalization

The Intuition-Sensing axis can be expressed via a free-energy-style expression:

$$F^{(N-S)} = D[Q^i(s^i) \parallel P^i(s^i \mid o^i)] - \mathbb{E}[\ln P^i(o^i)]. \quad (8)$$

Where:

- $Q^i(s^i)$: “Intuitive” predictions (N).
- $P^i(s^i \mid o^i)$: Data-driven (S) posterior or likelihood given observations o^i .
- $\mathbb{E}[\ln P^i(o^i)]$: Expected log-likelihood of observations, approximating how well o^i is explained.

5.2.2 T-F Axis Formalization

The Thinking-Feeling axis pairs are represented through specialized free-energy formulations:

1. Ti-Fe Pairing:

$$F_{\text{Ti-Fe}}^i = D[Q_{\text{refined}}^i(e) \parallel P^i(e \mid a_{\text{communal}}^i)]. \quad (9)$$

Where $Q_{\text{refined}}^i(e)$ represents precise internal/logical models.

2. Te-Fi Pairing:

$$F_{\text{Te-Fi}}^i = D[Q_{\text{adaptive}}^i(e) \parallel P^i(e \mid a_{\text{intrinsic}}^i)]. \quad (10)$$

Where $Q_{\text{adaptive}}^i(e)$ represents flexible external/logical models.

Note: The above is one possible stylized expression focusing on the KL divergence term. In a more typical Active Inference approach, the free energy might also include terms for $\mathbb{E}_{Q^i}[\ln p(o^i \mid s^i)]$ or $H[Q^i(s^i)]$. We keep it abstract for clarity in describing how certain personality axes might revolve around specialized divergences.

5.3 Hierarchical Processing and Temporal Dynamics

5.3.1 Markov Chain Representation

CompPsycho employs Markov chains to model transitions between cognitive states:

$$P(X_{t+1} = j \mid X_t = i) = p_{ij} \quad (11)$$

Where:

- X_t represents the cognitive state at time t
- p_{ij} represents the transition probability from state i to state j

The transition matrix $P = [p_{ij}]$ characterizes an individual’s personality type through their typical state transitions.

	Temporal Levels		
	Short-term	Medium-term	Long-term
High-Level	$B_{H,S}$	$B_{H,M}$	$B_{H,L}$
Context-Specific	$B_{C,S}$	$B_{C,M}$	$B_{C,L}$
Task-Specific	$B_{T,S}$	$B_{T,M}$	$B_{T,L}$

Table 1: Block matrix representation of behavioral motivations across temporal and contextual levels, where $B_{i,j}$ represents the behavioral dynamics at the intersection of contextual level i and temporal level j .

5.3.2 Extended Dynamic Modeling

The framework can be adapted to incorporate multiple temporal levels crossed with contextual levels in the definition of behavior. This creates a 3×3 block matrix structure:

Where:

- Temporal Levels:
 - Short-term (S): moment-to-moment behavioral motivations
 - Medium-term (M): weekly/monthly behavioral patterns
 - Long-term (L): year-scale behavioral dispositions
- Contextual Levels:
 - High-Level (H): overarching life themes and values
 - Context-Specific (C): domain-specific motivations
 - Task-Specific (T): immediate situational motivations

5.4 Rationale for a Two-Dimensional Matrix of Time and Context Levels

5.4.1 Integrating Temporal and Contextual Dimensions

In extending Active Inference to a richer personality and motivational model, we combine temporal scales (short, medium, long) with contextual levels (high-level, context-specific, task-specific) into a 2D block matrix. Each cell (or “block”) in this matrix defines a time-inhomogeneous Markov chain describing transitions among behavioral or cognitive states for that specific combination of timescale and context. Below, we articulate how these categories draw on well-established theories:

1. Time Dimension (Short, Medium, Long).

- Inspired by multi-level personality frameworks (e.g., Revelle & Condon, 2014), we recognize that behaviors manifest differently over moments (short-term states), weeks/months (medium-term patterns), and years (long-term traits).
- This resonates with dynamic models of personality (Atkinson & Birch, 1970; Fleeson, 2007) in which short-term fluctuations can accumulate into medium-term routines and eventually form long-term dispositions.

2. Context Dimension (High-Level, Context-Specific, Task-Specific).

- Borrowing from hierarchical motivation theories, especially Vallerand’s (1997) three-level model (Global, Contextual, Situational), we propose:
- High-Level: Overarching life themes or values, akin to Vallerand’s “Global” or Kruglanski et al.’s (2002) superordinate goals.
- Context-Specific: Domain or situation-level motivations (e.g., work, family, social roles). This maps onto Vallerand’s “Contextual” scale or “life-domain” focus in Self-Determination Theory (Deci & Ryan, 2000).

- Task-Specific: Immediate, moment-to-moment or “Situational” motivations, closely paralleling Vallerand’s “situational” and Action Identification Theory’s (Vallacher & Wegner, 1987) fine-grained level of action details.
- These three context tiers also align with the notion that goals, actions, or meanings can be framed at different levels of abstraction (Carver & Scheier, 1982; Vallacher & Wegner, 1987).

Hence, each dimension is rooted in existing frameworks—Revelle & Condon’s multi-level timescale approach plus Vallerand’s, Deci & Ryan’s, and Kruglanski’s hierarchical motivation theories—for why it makes sense to demarcate (a) short vs. medium vs. long timescales and (b) high-level vs. context-specific vs. task-specific motivations.

5.4.2 Constructing the 2D Block Matrix

1. Each Cell as a Time-Inhomogeneous Markov Chain

- In each cell (e.g., “Short-term, High-level”), we define transitions among relevant states (motivational, cognitive, or affective) with probabilities that may evolve over time (hence “time-inhomogeneous”).
- For instance, a “Short-term, High-level” chain might capture fast changes in how an agent interprets an overarching life goal during daily fluctuations (e.g., renewed sense of meaning each morning).
- By contrast, a “Long-term, Task-specific” chain might capture gradual changes in how a specific professional task is approached over years of skill development.

2. Illustrative Example

- Short-term, Task-specific: An agent deciding how to solve a puzzle right now, with local free-energy deltas reflecting moment-by-moment action selection.
- Medium-term, Context-specific: The same agent’s transitions across typical weekly routines at work or home—still context-defined but aggregated over weeks.
- Long-term, High-level: The agent’s broader life themes (e.g., desire for career success, altruism, or personal growth) evolving over decades.

3. Utility of Each “Block”

- Segmenting the model prevents conflating immediate tasks with large-scale life goals or ephemeral changes with deep-seated traits.
- The block matrix design ensures that each timescale and context-tier gets a dedicated Markov chain representation, capturing distinct motivational dynamics.

5.4.3 Why This Integration Is Valuable

1. Bridging Micro and Macro Perspectives

- Many personality models focus either on short-term micro-processes (e.g., dynamic fluctuations in daily states) or long-term traits.
- Our two-axis approach (Time $\times$ Context) ensures we can observe how immediate, domain-specific, or overarching-level motivations interplay across short, medium, or long timespans (cf. Vallerand, 1997; Revelle & Condon, 2014).

2. Empirical Anchoring

- While each cell is theoretically derived from well-cited frameworks, the approach as a whole allows empirical testing: for instance, collecting data at multiple timescales (daily diaries vs. monthly check-ins vs. lifetime achievements) and across multiple contexts (work vs. family vs. personal hobbies).

- Ultimately, the 2D matrix invites multilevel and multi-context methods, consistent with hierarchical motivation theories and dynamic personality research.

3. Future Extensions

- Some authors (e.g., Kruglanski et al., 2002) propose a richer goal hierarchy, while Revelle & Condon highlight multiple timescales. Our approach can be adapted if future empirical studies show that a 4-level or 5-level partitioning fits better.
- For now, the 3×3 matrix strikes a balance between theoretical grounding (Vallerand’s three-level model + short/medium/long times) and practical model complexity.

In sum, we adopt a 2D block matrix to systematically intersect (a) three timescales (short, medium, long) with (b) three context categories (high-level, context-specific, task-specific). Each of the resulting nine blocks captures distinct Markovian dynamics. This structure has roots in hierarchical motivation models (Vallerand, Action Identification, Self-Determination) and multi-level time frameworks (Revelle & Condon), providing a coherent, literature-backed basis for modeling the interplay between short-term local free-energy updates and broader, context-dependent personality processes.

5.5 Agent Stochastic Models and Transfer Entropy

Despite partitioning each agent’s behavior into multiple time-inhomogeneous Markov chains—one for each combination of temporal scale and contextual level—the overall framework still preserves stochastic modeling. Specifically:

1. Time-Inhomogeneous Markov Chains

- Each agent’s behavior at a given timescale (short-, medium-, or long-term) and context (high-level, context-specific, or task-specific) is encoded via transition probabilities that vary with time and/or contextual conditions.
- This retains the core property of a Markov process, namely that transitions out of a current state depend on a well-defined probability distribution (albeit evolving over time).

2. Stochastic Nature at Multiple Levels

- Because we do not impose deterministic transitions, each agent’s actions at each level remain probabilistic. The matrix of transition probabilities for, say, “short-term, high-level” behaviors may differ from that for “long-term, task-specific” behaviors but, in all cases, the agent’s next state is determined by probabilistic rules.
- This ensures each agent’s entire set of Markov chains (one per temporal-context block) collectively defines a family of stochastic processes.

3. Transfer Entropy Computation

- The transfer entropy between any two Markov processes tracks how much knowledge of one process reduces uncertainty about the future states of another.
- Given our block-matrix approach, we can compute transfer entropy:
 - (a) *Within an agent*, between different temporal or contextual chains (e.g., how “long-term, high-level” states influence subsequent “short-term, task-specific” states).
 - (b) *Across agents*, between corresponding or different blocks (e.g., how Agent A’s “medium-term, context-specific” chain influences Agent B’s “short-term, context-specific” chain).
- As each block remains a well-defined stochastic process with discrete (or continuous) time steps, transfer entropy can be estimated using standard information-theoretic methods on paired time-series data from the respective Markov chains.

4. Practical Advantages

- Segmenting each agent’s behavior into multiple Markov chains not only preserves stochastic integrity but also enables fine-grained analysis of information flow.
- Researchers can pinpoint exactly which level (temporal-context) exerts predictive power over another chain, thereby elucidating how local actions or beliefs propagate across timescales, contexts, and agents.

In summary, each agent remains a stochastic model comprising a suite of time-inhomogeneous Markov chains across temporal-contextual blocks. These chains support standard transfer entropy computations, permitting rigorous assessment of information transfer both within and between agents at each definable level of behavior.

5.6 Personality Type Formalization

5.6.1 State Space Definition

$$S = \{s_i | s_i = f(CV_i, H_i, F_i)\} \quad (12)$$

Where:

- CV_i : Cognitive vectors (Animals)
- H_i : Entropy states
- F_i : Free energy states

5.6.2 Transition Probability Space

The complete personality type space T is defined as:

$$T = \{P | P \in \mathbb{R}^{16 \times 16}, \sum_j p_{ij} = 1\} \quad (13)$$

This creates a continuous spectrum of $16!$ possible base personality types, each representing a unique pattern of cognitive state transitions. One can increase the granularity making the state space larger and the possible base personality types larger.

5.7 Practical Implementation

5.7.1 Binary Classification Framework

CP implements these theoretical constructs through:

1. Binary token pairs representing cognitive axes
2. Linear separability in high-dimensional space
3. Hierarchical prototypical networks for classification

5.7.2 Computational Architecture

The framework allows one to utilize statistical learning for:

1. State classification
2. Complex pattern recognition
3. Temporal dependency modeling

This formalization provides a mathematical foundation for understanding personality and cognition while enabling practical implementation in computational systems.

6 Local Dynamics and Global Invariance

6.1 Local Free Energy Dynamics

Each agent’s local free energy trajectory follows:

$$dF_{local} = \sum_{i=1}^n \Delta F_i(t) dt + \sigma(t) dW_t$$

where:

- $\Delta F_i(t)$ represents local free energy deltas
- $\sigma(t)$ is time-varying volatility
- W_t is a Wiener process capturing local fluctuations

Due to computational irreducibility:

- Local trajectories cannot be explicitly computed
- Exhibit stochastic variations
- Allow for temporary increases in free energy

6.2 Global Invariance

Despite local variations, global free energy minimization is preserved:

$$\mathbb{E} \left[\int_0^T dF_{global} \right] < 0$$

This creates a structure similar to a convolutional operator:

- Locally varying but globally invariant
- Allows local fluctuations while maintaining global properties
- Preserves minimization principle across scales

Conceptual Takeaway

Even though individuals may experience unpredictable shifts in their mental or behavioral states, the overall system tends to move toward stability over time. This means that while local fluctuations occur, a larger guiding principle ensures coherence at a broader level—much like how momentary thoughts and emotions can vary, yet underlying cognitive processes maintain overall balance and adaptation.

7 Information Preservation

7.1 Local-Global Information Content

The mutual information between global and local dynamics is preserved:

$$I(F_{global}; F_{local}) = H(F_{global}) - H(F_{global} | F_{local}) > 0$$

This ensures:

- Global minimization information exists in local trajectories
- Local behavior contains necessary global constraints
- Information about global principles is never lost

7.2 Stochastic Model Justification

The computational irreducibility and information preservation justify:

1. Time-inhomogeneous Markov chain representation:

$$P(\Delta F_{t+1}|F_t, F_{t-1}, \dots) = P(\Delta F_{t+1}|F_t)$$

2. Local martingale property:

$$\mathbb{E}[F_{t+1}|F_t, \dots, F_0] \leq F_t$$

3. Preservation of global structure despite local variance:

$$I(F_{global}; \{F_t\}_{t=0}^T) > 0$$

8 Foundational Principles

8.1 Core Properties

This framework establishes:

1. Local computational irreducibility with global constraints
2. Information preservation across scales
3. Justification for stochastic modeling
4. Compatibility of local variation with global principles

8.2 Implications for Modeling

These properties necessitate:

1. State space that preserves global information
2. Categories that support stochastic behavior
3. Structure that maintains predictive power
4. Framework for influence while preserving autonomy

9 Main Framework: Behavioral Space Partitioning

9.1 Abstract

We present a unified mathematical framework that demonstrates how Steiner systems and hypersphere packing provide an optimal schema for stochastic behavioral modeling, and prove why this structure must necessarily be present in language that effectively communicates behavior.

10 Fundamental Requirements for Behavioral Modeling

10.1 Core Constraints

For any behavioral modeling schema, the following properties must hold:

1. **Mutual Exclusivity:**

$$\forall b \in B, \exists! c \in C : b \in c$$

where B is the set of behaviors and C is the set of categories

2. **Collective Exhaustiveness:**

$$\bigcup_{c \in C} c = B$$

3. **Optimal Category Count:**

- Sufficient granularity: $|C| \geq \log_2(|B|)$
- Manageable complexity: $|C| \leq O(|B|^{1/2})$

4. **Uniform Average Probability:**

$$\forall c_i, c_j \in C : \mathbb{E}_A[P(c_i)] \approx \mathbb{E}_A[P(c_j)]$$

where $\mathbb{E}_A$ is expectation over agents

10.2 State Sufficiency Requirements

The schema must support:

1. Markov property: $P(s_{t+1}|s_t, s_{t-1}, \dots) = P(s_{t+1}|s_t)$
2. Higher-order dependencies through state structure
3. Natural symmetries justifying state transitions

11 Steiner System Framework

11.1 Basic Structure

For Steiner system $S(t, k, v)$:

- v elements represent fundamental behavioral components
- k size blocks represent categorical groupings
- t -wise intersections capture behavioral dependencies

Number of blocks:

$$b = \frac{\binom{v}{t}}{\binom{k}{t}}$$

11.2 Behavioral Mapping Properties

Each block B_i represents a behavioral category with:

1. Internal coherence through k -subset structure
2. Controlled overlap through t -wise intersections
3. Natural hierarchical structure through subset relationships

12 Hypersphere Packing Correspondence

12.1 Geometric Embedding

For each block B_i , define corresponding hypersphere:

$$H_i = \{x \in \mathbb{R}^d : \|x - c_i\| \leq r_i\}$$

where:

- c_i is the category centroid
- r_i is the categorical radius
- d is optimal embedding dimension

12.2 Optimal Packing Properties

For hyperspheres H_i, H_j :

1. Minimal overlap: $\text{vol}(H_i \cap H_j) \leq \epsilon$
2. Maximal coverage: $\text{vol}(\bigcup_i H_i) \approx \text{vol}(B)$
3. Uniform distribution: $\text{vol}(H_i) \approx \text{vol}(H_j)$

13 Prototype Class Correspondence

13.1 Formal Correspondence

For n prototype classes $P = \{P_1, \dots, P_n\}$, optimal hypersphere packing determines minimal dimensionality preserving distinct relationships.

Let μ_i be centroid of class P_i

Let Σ_i be covariance matrix

For separation ϵ :

$$\|\mu_i - \mu_j\| \geq \epsilon + r_i + r_j$$

where $r_i = \sqrt{\text{tr}(\Sigma_i)}$

Optimal dimension d^* satisfies:

1. Maximizes packing density $\eta(d)$
2. Minimizes dimension d
3. Maintains separation ϵ

13.2 State Space Properties for Communication

The Steiner system blocks as states enable:

1. Efficient binary discrimination through exclusion
2. Natural modification of transition probabilities
3. Preservation of essential behavioral structure

For linguistic signals L modifying transitions:

$$P_{ij}(t+1) = P_{ij}(t) \cdot f(L, B_i, B_j)$$

where f preserves:

- Markov property
- Free energy minimization
- Computational irreducibility

13.3 Optimality Properties

1. Minimal embedding dimension:

$$d^* \geq \log_2(n) + O(1)$$

2. Optimal separation:

$$d_H(B_i, B_j) = \text{constant}$$

3. Natural class boundaries through sphere geometry

14 Effective Communication and Information Transfer

14.1 Effective Communicator Definition

An agent A is an "effective communicator" if it can generate linguistic signals L that:

1. Intentionally influence another agent B 's behavioral trajectory
2. Increase transfer entropy between chains:

$$T_{A \rightarrow B} = I(L; \Delta F_B(t+1) | \Delta F_B(t)) > 0$$

3. Achieve this with minimal encoding length $H(L)$

14.2 Minimal Effective Linguistic Signal

The minimal information content required for an effective linguistic signal must include at least one block exclusion from the Steiner system.

1. Let M_A, M_B be time-inhomogeneous Markov chains for agents A, B
2. For effective communication:

$$T_{A \rightarrow B} = I(L; \Delta F_B(t+1) | \Delta F_B(t))$$

must be achieved with minimal $H(L)$

3. Consider signal $L(t) = \{(B_{exclude}, \Delta)\}$:
 - Modifies M_B 's transition probabilities
 - $H(L) = \log_2(b-1) + H(\Delta)$
 - No smaller encoding can guarantee $T_{A \rightarrow B} > 0$
4. Block exclusion provides:
 - Binary discrimination power
 - Minimal information content
 - Guaranteed trajectory influence

14.3 Application to Language Analysis

This framework justifies:

1. Searching for block exclusion patterns in behavioral communication
2. Analyzing language through state space modifications
3. Identifying effective communication through minimal exclusions

14.4 Transfer Entropy Framework

Each agent's behavior follows a time-inhomogeneous Markov chain M_A where:

- States correspond to blocks B_i in Steiner system $S(t, k, v)$
- Transitions: $P(\Delta F_{t+1} = B_j | \Delta F_t = B_i) = P_{ij}(t)$
- Free energy deltas $\Delta F_i(t)$ define local dynamics

For communication between agents A and B , transfer entropy is:

$$T_{A \rightarrow B} = I(L; \Delta F_B(t+1) | \Delta F_B(t))$$

where L represents linguistic signals modifying B 's behavior.

14.5 Block Exclusion for Optimal Encoding

The most efficient encoding of linguistic signals corresponds to excluding at least one block from the agent's available state transitions.

1. Let $L(t) = \{(B_{exclude}, \Delta)\}$ be linguistic signal
2. Original entropy: $H(L) = \log_2(b(b-1)) + H(\Delta)$
3. With $m \geq 1$ blocks excluded:

$$H(L_m) = \log_2((b-m)(b-m-1)) + H(\Delta)$$

4. Due to t -wise intersections:
 - Any state representable by combinations in $(b-1)$ system
 - Additional blocks add redundant information
 - $H(L_m) < H(L)$ while maintaining $T_{A \rightarrow B}$

14.6 Optimality Properties

Block exclusion achieves:

1. Minimal encoding length: $H(L) = \log_2(b-1) + H(\Delta)$
2. Maximum influence per bit of information
3. Preservation of agent autonomy through:
 - Maintained computational irreducibility
 - Preserved stochastic dynamics
 - Retained free energy minimization

15 Stochastic Process Properties

15.1 Markov Chain Structure

Agent behavior follows:

$$P(\Delta F_{t+1} = B_j | \Delta F_t = B_i) = P_{ij}(t)$$

Justified by:

1. Geometric symmetries of packing
2. Natural state boundaries from spheres
3. t -wise intersections capturing dependencies

15.2 Information Theoretic Properties

1. Maximum entropy:

$$H(B) = - \sum_i P(B_i) \log P(B_i) \approx \log |C|$$

2. Transfer entropy optimization:

$$T_{A \rightarrow B} = I(L; \Delta F_B(t+1) | \Delta F_B(t))$$

3. Optimal encoding through block exclusion:

$$H(L) = \log_2(b-1) + H(\Delta)$$

16 Constraint Satisfaction Proof

The Steiner system with hypersphere packing satisfies all behavioral modeling constraints.

1. Mutual Exclusivity:
 - Follows from block structure
 - Reinforced by hypersphere boundaries
2. Collective Exhaustiveness:
 - Guaranteed by t -wise coverage
 - Optimized by packing density
3. Optimal Categories:
 - Block count b balances granularity and complexity
 - Dimension d^* provides natural scale
4. Uniform Probability:
 - Emerges from geometric uniformity
 - Preserved by packing properties
5. State Sufficiency:
 - Geometric symmetries justify Markov property
 - Block structure captures dependencies
 - Natural hierarchies enable prediction

17 Necessary Presence in Behavioral Language

17.1 From Optimal Structure to Linguistic Necessity

The Steiner system block structure must be present in any language that effectively communicates behavior.

1. From previous theorems:
 - Steiner system provides optimal behavioral categorization
 - Hypersphere packing gives natural prototype structure
 - Block exclusion is minimal requirement for effective communication

2. For any agent to be an "effective communicator":
 - Must achieve $T_{A \rightarrow B} > 0$
 - Must do so with minimal information
 - Proven to require at least block exclusion
3. Therefore, any linguistic signal L that effectively communicates behavior:
 - Must contain block exclusion information
 - Must reflect the underlying Steiner structure
 - Cannot achieve effective communication without this

17.2 Implications for Behavioral Language

This proves:

1. The Steiner block structure isn't just descriptive but necessary
2. Must be discoverable in effective behavioral communication
3. Provides a theoretical basis for analyzing behavioral language

18 Summary

The framework demonstrates:

1. Requirements for behavioral modeling $\rightarrow$ Steiner system optimal categorization $\rightarrow$ Hypersphere packing and prototype structure $\rightarrow$ Markov chain representation $\rightarrow$ Block exclusion as minimal effective signal $\rightarrow$ Necessary presence in behavioral language

This framework demonstrates that Steiner systems with hypersphere packing provide:

1. Optimal behavioral space partitioning
2. Natural prototype class structure
3. Justified stochastic modeling properties
4. Satisfaction of all core behavioral modeling constraints

19 Mathematical Framework Starting from a Steiner System

19.1 Steiner System Preliminaries

A Steiner system is a finite combinatorial design: a set of elements together with a collection of k -element subsets of X (called blocks), such that every t -element subset of X is contained in exactly one block. Here, consider the system $S(3, 4, 8)$:

In an $S(3, 4, 8)$, each triple of elements of X uniquely determines a single 4-subset (block) in $\mathcal{B}$.

19.2 Behavioral Categories as Blocks

Suppose we have four categories C_1, C_2, C_3, C_4 , each corresponding to a specific block in the Steiner system. Let these categories be represented by distinct 4-subsets of the 8-element set X :

Since each triple identifies a unique block, each category is combinatorially well-defined and distinct. Assume the categories correspond to a known symmetric configuration within $S(3, 4, 8)$.

19.3 Embedding into $\mathbb{R}^2$

We posit that the four categories, though combinatorial in nature, can be associated with points in a 2-dimensional vector space $\mathbb{R}^2$. This embedding is chosen so that each category differs from its “opposite” category in both coordinates. For instance, encode each category as follows:

This setup ensures that every pair of “opposite” categories—those differing in both underlying attributes—correspond to diametrically opposed points in $\mathbb{R}^2$.

Such an embedding may be constructed by identifying the attributes (e.g., Orientation and Decision) with dimensions of the vector space. The Steiner system structure ensures a high degree of symmetry, allowing for a natural linear representation of the categories.

19.4 Partial Information and Opposites

In practice, we do not always trust the full categorical assignment. Instead, from an uncertain guess that the behavior belongs to some category C_i , we extract partial information: it is at least not the opposite category $\bar{C}_i$. These “not-opposite” constraints correspond, in the Steiner system context, to excluding one block from consideration. Since each triple in $S(3, 4, 8)$ identifies exactly one block, removing that block eliminates the unique triple it covers, simplifying the classification problem.

Formally, if we pair C_1 with its opposite C_2 , and C_3 with its opposite C_4 , the partial information “not C_2 ” given a suggestion of C_1 translates to a linear inequality on $\mathbb{R}^2$, and similarly for the pair C_3, C_4 .

19.5 Defining New Axes from Partial Information

To incorporate this partial information into a coherent coordinate system, consider the vector differences defining each pair of opposites:

$$\vec{v}_1 = C_1 - C_2, \quad \vec{v}_2 = C_3 - C_4 \quad (14)$$

These vectors $\vec{v}_1$ and $\vec{v}_2$ represent contrasts derived from the partial information constraints “not the opposite category.” Since the Steiner system guarantees the uniqueness of blocks containing certain triples, these contrasts are not arbitrary—they align with symmetries inherent in the combinatorial design. Each “not-opposite” rule corresponds to removing one block, which under the design is a unique combinatorial event. Thus, the direction $\vec{v}_1$ encodes a dimension of guaranteed distinctiveness (C_1 vs. C_2), and $\vec{v}_2$ encodes another (C_3 vs. C_4).

Choose a normalized or simplified basis from these vectors, for instance:

$$\vec{e}_1 = \frac{\vec{v}_1}{|\vec{v}_1|}, \quad \vec{e}_2 = \frac{\vec{v}_2}{|\vec{v}_2|} \quad (15)$$

Construct the linear transformation T with $\vec{e}_1, \vec{e}_2$ as the rows:

$$T = \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \end{bmatrix} \quad (16)$$

Applying T to a category yields new coordinates:

$$T(C_i) = \begin{bmatrix} \langle \vec{e}_1, C_i \rangle \\ \langle \vec{e}_2, C_i \rangle \end{bmatrix} \quad (17)$$

Under this mapping:

- Each category aligns neatly along one axis or the other
- The partial-information constraints appear as sign conditions on x_1 or x_2

19.6 Relation to Steiner System Symmetries

The Steiner system is known for its rich symmetry group. Automorphisms of the system permute elements of X and blocks in $\mathcal{B}$ while preserving the uniqueness property of triples. Such symmetries correspond to transformations in the associated vector space representation.

The matrix T can be regarded as an algebraic avatar of an automorphism that reorients the combinational structure to a frame aligned with the partial information constraints. By starting from the premise that categories are embedded in $\mathbb{R}^2$, we ensure that each partial-information axis (like “not the opposite block”) corresponds to a well-defined and unique structural property of the design. Thus, the linear transformation is not arbitrary; it is grounded in the Steiner system’s combinatorial guarantee that excluding one block (identified by a triple) isolates a unique set of remaining options.

20 Integrating Partial Information Through a Steiner System Framework

In many behavior classification paradigms, complex decision-making is reduced to identifying a subject’s actions within a small set of well-defined categories. Consider a scenario wherein each behavior is classified according to two binary attributes—such as “Orientation” (O) and “Decision” (D)—each taking one of two possible values. For example, Orientation might be Extroverted (O_e) or Introverted (O_i), and Decision might be Extroverted (D_e) or Introverted (D_i). Combining these two binary dimensions yields four distinct categories.

Geometrically, these four points can be represented as in a 2-dimensional (O, D) -plane, forming a square with vertices at:

$$(O_e, D_e), (O_i, D_i), (O_e, D_i), (O_i, D_e) \quad (18)$$

While this idealized representation is conceptually clear, real-world data acquisition often provides only partial or noisy information about a subject’s underlying processes. Observers may guess that a given behavior corresponds to a particular category (e.g., “This seems like $O_e D_e$ ”), yet this guess may not be fully reliable. Instead of accepting the guessed category outright, we focus on a key piece of partial information that can be trusted: if the guess suggests $O_e D_e$, at least we can be confident that the behavior is not $O_i D_i$, its “opposite” category differing on both binary attributes. Similarly, a guess leaning toward $O_i D_e$ allows us to conclude the behavior is not $O_e D_i$.

These “not-opposite” constraints translate uncertain guesses into partial yet reliable distinctions. Each such piece of information identifies a direction in which we can confidently differentiate categories. Specifically, from “ $O_e D_e$ is not $O_i D_i$,” we establish one axis of contrast, and from “ $O_i D_e$ is not $O_e D_i$,” we establish another. These two new contrasts serve as alternative coordinate axes, replacing the original (O, D) dimensions with two new dimensions.

This redefinition can be viewed mathematically as applying a linear transformation to the original coordinate system. Instead of directly measuring “Orientation” and “Decision,” we measure how much more “like $O_e D_e$ than $O_i D_i$ ” and “like $O_i D_e$ than $O_e D_i$ ” a behavior appears. By plotting behaviors in this transformed 2D space, each one is located according to partial, trustworthy differences, rather than uncertain absolute attributes.

20.1 Connection to the Steiner System

A powerful combinatorial structure underpins this reasoning: the Steiner system $S(3, 4, 8)$. A Steiner system is a set of elements with blocks of size k such that every subset of size t is contained in exactly one block. The system $S(3, 4, 8)$ is known for its remarkable symmetry and uniqueness properties, mapping every triple of elements to exactly one of several 4-element blocks. This exact covering property mirrors the resolution of partial information in our classification task. Just as ruling out one category’s opposite narrows our search to a unique structural solution, a Steiner system’s combinatorial precision ensures that excluding certain triples (analogous to excluding “opposite” configurations) isolates a unique block.

In more concrete terms, if we embed our four-category classification into an $S(3, 4, 8)$ design, each category corresponds to a distinguished 4-subset (block) of an 8-element universe. The condition that each triple

of elements identifies exactly one block ensures minimal ambiguity: any piece of partial information that removes one block (i.e., rules out a category or its opposite) leads to a unique determination of the remaining possibilities. Thus, identifying new axes based on “not-opposite” constraints corresponds to employing the Steiner system’s combinational structure. The redefined axes leverage the system’s property that any partial exclusion of one block’s defining subset leaves a configuration still positioned uniquely among the others.

20.2 Synthesis and Implications

This connection to a Steiner system provides a robust theoretical foundation for our partial-information approach. Rather than treating partial information heuristically, we see that every “not-opposite” deduction corresponds to a well-defined combinatorial constraint. The Steiner system’s stringent coverage conditions ensure that combining such constraints leads to a coherent coordinate system. As a result, the transformed axes—derived purely from partial, reliable distinctions—still form a stable classification framework. The four categories remain distinct and can be identified through a series of simple, trustable negations, each corresponding to known symmetries and unique coverings within the Steiner system.

In summary, the linkage of our binary classification model to a Steiner system reveals why partial information, embodied in “not the opposite category” constraints, suffices to construct meaningful axes of distinction. The underlying combinatorial design ensures that every partial piece of information reliably enhances resolution, and the resulting coordinate system naturally maintains the structural integrity of the original four-category classification—albeit in a reoriented, data-driven fashion.

21 From the Free Energy Principle to a Markov-Chain Approximation

In this section, we provide the necessary background on the Free Energy Principle (FEP) and its interpretation as a principle of least action, before introducing our Markov-chain-based approximation for capturing free-energy trajectories (§23). Our treatment follows recent expositions in FristonFEP2023, Friston2019FEP, DaCosta2021BMech and references therein.

21.1 Active Inference and the Free Energy Principle as Least Action

Active Inference and Self-Evidencing. Active Inference ? extends the long-standing idea that biological systems (e.g. brains, organisms) can be modeled as *Bayesian* or *variational* learners, predicting incoming sensory data and updating internal states to minimize prediction error. In more physical terms, such inference and learning can be cast as *minimizing a free-energy functional*. Crucially, this free energy can be interpreted as an *action functional* in a path-integral sense (?, §7–8), akin to Hamilton’s principle of least action in classical mechanics.

Principle of Least Action in FEP. Formally, starting with a stochastic description of a system’s states $x(\tau)$ under random fluctuations, one obtains a *path-integral* expression for the *action* $\mathcal{A}[x(\tau)]$. In the small-fluctuation (or “precise particle”) regime, the system’s *most likely* path $\mathbf{x}[\tau]$ is precisely the path of least action:

$$\mathbf{x}[\tau] = \arg \min_{x[\tau]} \mathcal{A}[x(\tau)]. \quad (19)$$

Under the Free Energy Principle, the same *least-action* path emerges by considering *variational free energy* ???. Hence, the FEP essentially states that *self-organizing systems* will adopt trajectories that locally (and often globally) minimize a free-energy functional; one can therefore read $\mathcal{A}[x(\tau)]$ as the negative log-probability (or “surprisal”) of the path, such that minimizing action is tantamount to minimizing surprise or free energy ??.

Interpretation for Behavioral or Cognitive Trajectories. In contexts such as *Active Inference*, states $x(\tau)$ often include an agent’s *internal states* (encoding beliefs) and *blanket states* (active + sensory). The

path that *reduces* variational free energy over time is that which *best* explains sensory data while satisfying prior (Bayesian) constraints ??.

21.2 Rationale for a Markov-Chain Approximation of Free-Energy Trajectories

Although a continuous-time path-integral formulation is elegantly general, it may be *computationally* demanding for real systems. Empirical or simulation-based research often deals with discrete time steps $t = 0, 1, \dots, N$, and observes *short-run changes* in free energy $F(t)$. In many practical scenarios:

- We only have discrete snapshots of $F(t)$ or approximate *increments* $\Delta F_t = F(t+1) - F(t)$,
- The agent’s policy or environmental influences *evolve* over time, leading to *nonstationary* or *time-varying* dynamics,
- We may want to simulate *possible* next steps, or forecast how $F(t)$ might evolve under a new policy.

Hence, it is natural to seek a *Markov-chain* model—particularly a *time-inhomogeneous* Markov chain—on these free-energy increments or deltas.

Link to Least-Action / Most-Likely Path. In Section 23, we show that fitting a discrete-time Markov chain to observed free-energy changes *recovers* both:

- (i) *short-term* rises or fluctuations (which appear as states/transitions of $\Delta F_t > 0$),
- (ii) *the overall least-action or free-energy-reducing path* (which, in a low-noise regime, the chain will favor with high probability).

Accordingly, the chain becomes a data-driven, coarse-grained approximation to the system’s “free-energy trajectory.” If ΔF_t are measured or *inferred* from partial observations, the chain can replicate the *empirical shape* of $F(t)$ with sufficient data.

Time-Inhomogeneity and Changing Behaviors. Because real agents often *adapt* their policies or respond to new conditions, the transition probabilities cannot be strictly stationary. Instead, we allow $P_t(X_{t+1} | X_t)$ to change with t —reflecting the agent’s updated behavioral dispositions (e.g., after learning) or external interventions. This aligns with the *dynamic* concept of Active Inference (where agents continually refine internal parameters ??).

Preview. Section 23 below details how one can implement, estimate, and justify this *Markov-chain approximation to free-energy deltas*, proving that it converges (with large data) to an accurate representation of the system’s actual free-energy path. In particular, we see how:

- *Short-run* increments or “bumps” of free energy are captured as transitions to positive- ΔF states.
- *Long-run* “least-action” minimization emerges from the chain’s empirical transition structure, which heavily weights *negative- ΔF* transitions (in typical FEP scenarios).
- *Time-inhomogeneity* allows for the chain to encode the *evolving*, context-dependent nature of an agent’s free-energy trajectory—thus capturing the continuous changes in policy or environment that shape how $F(t)$ evolves.

This approach both *operationalizes* the free energy principle for empirical data analysis and provides a simpler discrete-time tool for modeling *behavioral evolution*, consistent with the underlying principle of least action.

22 A Markov-Chain Approximation of Free-Energy Deltas and Behavioral Evolution

This section develops a formal argument that a Markov chain whose state space comprises *free-energy deltas* (increments or decrements of a global free-energy measure) can, under suitable data availability, converge to an empirically valid approximation of the system’s actual free-energy trajectory. Moreover, we show how a *time-inhomogeneous* version of this Markov chain can encode the agent’s changing behavioral tendencies or policies over time, including possible external influences.

22.1 Setup: Tracking Free-Energy Deltas

Let $\{F(t)\}_{t=0}^N$ denote a discrete-time record (empirical or simulated) of an agent’s free energy at integer times $t = 0, 1, \dots, N$. Define the *free-energy increment* (or delta)

$$\Delta F_t = F(t+1) - F(t), \quad \text{for } t = 0, 1, \dots, N-1.$$

Each ΔF_t captures the local change in free energy between successive time steps. In many behavioral models under the Free Energy Principle (FEP), these changes can reflect (i) short-term *increases* in free energy for exploration or barrier-crossing, and (ii) more prevalent *decreases* consistent with overall free-energy minimization ??.

Markov States as Free-Energy Increments

We consider a (finite or countably infinite) state space $\mathcal{X}$ partitioning or categorizing the possible values of ΔF_t . For example, one might discretize ΔF_t into bins (e.g., ‘‘Large Positive’’, ‘‘Slight Positive’’, ‘‘Slight Negative’’, ‘‘Large Negative’’), or equivalently track absolute levels $\{F(t)\}$ as states. Let

$$X_t \approx \Delta F_t \quad (\text{or } F(t))$$

be the Markov chain’s state at time t . The chain will attempt to capture how ΔF_{t+1} depends on ΔF_t (and possibly earlier increments, if extended to a higher-order chain).

22.2 Approximate Predictive Validity with Sufficient Data

Estimation and Convergence. Given a sufficiently large data set of length N , we collect pairs (X_t, X_{t+1}) empirically and estimate transition probabilities

$$\hat{P}_t(x, x') = \frac{\#\{\text{transitions } (x \rightarrow x') \text{ at time } t\}}{\#\{X_t = x\}}.$$

When $N \rightarrow \infty$ (and mild ergodic or stability conditions hold for each time interval), classical results in Markov-process and time-series analysis ?? imply that $\hat{P}_t \rightarrow P_t$ with high probability. Here, $P_t(x, x')$ denotes the *true* transition probability from x to x' at time t in the real system (if such a well-defined conditional distribution exists).

Predicting the Free-Energy Path. Once we have an estimated chain $\{\hat{P}_t\}_{t=0}^{N-1}$, we can *simulate* a synthetic path $\{\tilde{X}_t\}$ by drawing $\tilde{X}_{t+1}$ from $\hat{P}_t(\tilde{X}_t, \cdot)$. If we model ΔF_t as X_t , then reconstructing the free energy is done via:

$$\tilde{F}(0) = F(0), \quad \tilde{F}(t+1) = \tilde{F}(t) + \tilde{X}_t.$$

When $\hat{P}_t \approx P_t$, the distribution of $\{\tilde{F}(t)\}$ approximates that of the true $\{F(t)\}$: short-term increases and decreases appear at the correct frequencies, and the cumulative effect captures the empirically observed *shape* of $F(t)$ over time. This implies *approximate predictive validity*: the chain forecast for $F(t)$ remains close to the real trajectory, at least on average.

Including Deviations and an Optimal Path

Even if the system occasionally invests in *increases* in free energy (e.g. exploration or barrier crossing), these increments are reflected in the empirical data. As $\hat{P}_t \rightarrow P_t$, the Markov chain will place a correct probability mass on such *temporary rises*, while still (in a low-noise or large-sample regime) assigning higher probability to an overall *least-action* or *free-energy-reducing* trajectory. In other words, both the short-run fluctuations and the long-run minimal-action tendency are embedded in the transition probabilities.

22.3 Time-Inhomogeneous Transitions for Changing Behavioral Tendencies

In realistic settings, an agent’s motivations or policies may *evolve* over time. This yields a *time-inhomogeneous* Markov chain:

$$P_t(x, x') = \mathbb{P}[X_{t+1} = x' \mid X_t = x], \quad \text{where } P_t \text{ depends on } t.$$

Such inhomogeneity reflects changes in the agent’s internal states (learning, strategy shifts) or external manipulations (environmental or experimental influences). For example, P_t might remain approximately constant for $t_1 \leq t < t_2$ (a policy phase) and then abruptly switch to a new matrix P'_t for $t \geq t_2$ when a new influence is introduced.

Encoding the Evolution of Policies. In each sub-interval, the *chain structure* captures how ΔF_t transitions are distributed under the current policy or influence. Hence, a piecewise or smoothly time-varying $\{P_t\}$ can represent *policy adaptation*, shifting the likelihood of short-term $\Delta F > 0$ vs. $\Delta F < 0$. Over multiple intervals, the resulting chain encodes the entire *chronology* of how the agent’s tendency to raise or lower free energy changes in response to new goals, constraints, or learning signals.

Convergence in Nonstationary Settings. While stationarity no longer applies, standard theorems for *time-inhomogeneous* Markov chains ? establish that, under mild regularity, one can still estimate P_t from observed transitions within each phase (or within short windows of t). The estimated chain $\{\hat{P}_t\}$ again converges to $\{P_t\}$, and simulations from $\{\hat{P}_t\}$ replicate the piecewise or fully nonstationary behavior of ΔF_t over time.

22.4 Conclusion

We have shown that modeling *free-energy deltas* ΔF_t as states of a (time-inhomogeneous) Markov chain provides a feasible, data-driven approach to capturing:

- **Short-term rises and drops** in free energy that reflect momentary deviations, explorations, or barrier crossings.
- **Long-term or overall minimization** consistent with the Free Energy Principle’s least-action path in low-noise regimes.
- **Evolution of the agent’s behavioral tendency** when policy or external conditions shift, by allowing time-varying transition matrices P_t .

Provided the sample size N is large enough and mild ergodic or consistency conditions hold, the empirical chain $\{\hat{P}_t\}$ converges to the real dynamics of $\{\Delta F_t\}$. Thus, the chain achieves *approximate predictive validity*: it can forecast free-energy paths and highlight the agent’s shifting motivations over time. Consequently, a coarse-grained Markov chain over ΔF becomes a powerful tool for *explaining* and *predicting* the observed trajectory of free energy, without requiring direct knowledge of the underlying multidimensional state of the agent’s internal processes.

23 A Markov-Chain Approximation of Free-Energy Deltas and Behavioral Evolution

This section develops a formal argument that a Markov chain whose state space comprises *free-energy deltas* (increments or decrements of a global free-energy measure) can, under suitable data availability, converge to an empirically valid approximation of the system’s actual free-energy trajectory. Moreover, we show how a *time-inhomogeneous* version of this Markov chain can encode the agent’s changing behavioral tendencies or policies over time, including possible external influences.

23.1 Setup: Tracking Free-Energy Deltas

Let $\{F(t)\}_{t=0}^N$ denote a discrete-time record (empirical or simulated) of an agent’s free energy at integer times $t = 0, 1, \dots, N$. Define the *free-energy increment* (or delta)

$$\Delta F_t = F(t+1) - F(t), \quad \text{for } t = 0, 1, \dots, N-1.$$

Each ΔF_t captures the local change in free energy between successive time steps. In many behavioral models under the Free Energy Principle (FEP), these changes can reflect (i) short-term *increases* in free energy for exploration or barrier-crossing, and (ii) more prevalent *decreases* consistent with overall free-energy minimization ??.

Markov States as Free-Energy Increments

We consider a (finite or countably infinite) state space $\mathcal{X}$ partitioning or categorizing the possible values of ΔF_t . For example, one might discretize ΔF_t into bins (e.g., ‘‘Large Positive’’, ‘‘Slight Positive’’, ‘‘Slight Negative’’, ‘‘Large Negative’’), or equivalently track absolute levels $\{F(t)\}$ as states. Let

$$X_t \approx \Delta F_t \quad (\text{or } F(t))$$

be the Markov chain’s state at time t . The chain will attempt to capture how ΔF_{t+1} depends on ΔF_t (and possibly earlier increments, if extended to a higher-order chain).

23.2 Approximate Predictive Validity with Sufficient Data

Estimation and Convergence. Given a sufficiently large data set of length N , we collect pairs (X_t, X_{t+1}) empirically and estimate transition probabilities

$$\hat{P}_t(x, x') = \frac{\#\{\text{transitions } (x \rightarrow x') \text{ at time } t\}}{\#\{X_t = x\}}.$$

When $N \rightarrow \infty$ (and mild ergodic or stability conditions hold for each time interval), classical results in Markov-process and time-series analysis ?? imply that $\hat{P}_t \rightarrow P_t$ with high probability. Here, $P_t(x, x')$ denotes the *true* transition probability from x to x' at time t in the real system (if such a well-defined conditional distribution exists).

Predicting the Free-Energy Path. Once we have an estimated chain $\{\hat{P}_t\}_{t=0}^{N-1}$, we can *simulate* a synthetic path $\{\tilde{X}_t\}$ by drawing $\tilde{X}_{t+1}$ from $\hat{P}_t(\tilde{X}_t, \cdot)$. If we model ΔF_t as X_t , then reconstructing the free energy is done via:

$$\tilde{F}(0) = F(0), \quad \tilde{F}(t+1) = \tilde{F}(t) + \tilde{X}_t.$$

When $\hat{P}_t \approx P_t$, the distribution of $\{\tilde{F}(t)\}$ approximates that of the true $\{F(t)\}$: short-term increases and decreases appear at the correct frequencies, and the cumulative effect captures the empirically observed *shape* of $F(t)$ over time. This implies *approximate predictive validity*: the chain forecast for $F(t)$ remains close to the real trajectory, at least on average.

Including Deviations and an Optimal Path

Even if the system occasionally invests in *increases* in free energy (e.g. exploration or barrier crossing), these increments are reflected in the empirical data. As $\hat{P}_t \rightarrow P_t$, the Markov chain will place a correct probability mass on such *temporary rises*, while still (in a low-noise or large-sample regime) assigning higher probability to an overall *least-action* or *free-energy-reducing* trajectory. In other words, both the short-run fluctuations and the long-run minimal-action tendency are embedded in the transition probabilities.

23.3 Time-Inhomogeneous Transitions for Changing Behavioral Tendencies

In realistic settings, an agent’s motivations or policies may *evolve* over time. This yields a *time-inhomogeneous* Markov chain:

$$P_t(x, x') = \mathbb{P}[X_{t+1} = x' \mid X_t = x], \quad \text{where } P_t \text{ depends on } t.$$

Such inhomogeneity reflects changes in the agent’s internal states (learning, strategy shifts) or external manipulations (environmental or experimental influences). For example, P_t might remain approximately constant for $t_1 \leq t < t_2$ (a policy phase) and then abruptly switch to a new matrix P'_t for $t \geq t_2$ when a new influence is introduced.

Encoding the Evolution of Policies. In each sub-interval, the *chain structure* captures how ΔF_t transitions are distributed under the current policy or influence. Hence, a piecewise or smoothly time-varying $\{P_t\}$ can represent *policy adaptation*, shifting the likelihood of short-term $\Delta F > 0$ vs. $\Delta F < 0$. Over multiple intervals, the resulting chain encodes the entire *chronology* of how the agent’s tendency to raise or lower free energy changes in response to new goals, constraints, or learning signals.

Convergence in Nonstationary Settings. While stationarity no longer applies, standard theorems for *time-inhomogeneous* Markov chains ? establish that, under mild regularity, one can still estimate P_t from observed transitions within each phase (or within short windows of t). The estimated chain $\{\hat{P}_t\}$ again converges to $\{P_t\}$, and simulations from $\{\hat{P}_t\}$ replicate the piecewise or fully nonstationary behavior of ΔF_t over time.

23.4 Conclusion

We have shown that modeling *free-energy deltas* ΔF_t as states of a (time-inhomogeneous) Markov chain provides a feasible, data-driven approach to capturing:

- **Short-term rises and drops** in free energy that reflect momentary deviations, explorations, or barrier crossings.
- **Long-term or overall minimization** consistent with the Free Energy Principle’s least-action path in low-noise regimes.
- **Evolution of the agent’s behavioral tendency** when policy or external conditions shift, by allowing time-varying transition matrices P_t .

Provided the sample size N is large enough and mild ergodic or consistency conditions hold, the empirical chain $\{\hat{P}_t\}$ converges to the real dynamics of $\{\Delta F_t\}$. Thus, the chain achieves *approximate predictive validity*: it can forecast free-energy paths and highlight the agent’s shifting motivations over time. Consequently, a coarse-grained Markov chain over ΔF becomes a powerful tool for *explaining* and *predicting* the observed trajectory of free energy, without requiring direct knowledge of the underlying multidimensional state of the agent’s internal processes.

24 Time-Inhomogeneous Markov Chains and Transfer Entropy among Agents

In this section, we extend our Markov chain framework to multiple agents and explain how time-inhomogeneous chains can encode sufficient information to compute a *time-resolved* Transfer Entropy (TE), thereby tracking the causal or directed influence of one agent’s behavioral evolution on another.

24.1 Pairwise Transfer Entropy: A Quick Recap

Consider two discrete-time processes $\{X_t\}$ and $\{Y_t\}$, where $t \in \{0, 1, 2, \dots\}$. *Transfer Entropy* ?? from Y to X at time $t + 1$ is defined informally by:

$$T_{Y \rightarrow X}(t) = \sum_{x_{t+1}, x_t, y_t} p_t(x_{t+1}, x_t, y_t) \ln \left[\frac{p_t(x_{t+1} | x_t, y_t)}{p_t(x_{t+1} | x_t)} \right], \quad (20)$$

where $p_t(\cdot)$ denotes the (possibly time-varying) joint or conditional probabilities at time t . Intuitively, $T_{Y \rightarrow X}(t)$ measures how much knowing Y_t (in addition to X_t) improves the predictability of X_{t+1} . A larger TE indicates a stronger directed (or “causal-like”) influence from Y to X .

24.2 Modeling Behavioral Tendencies via Time-Inhomogeneous Chains

Behavioral Transitions for Multiple Agents. Suppose we have two agents, $\mathcal{A}$ and $\mathcal{B}$, each characterized by a (time-inhomogeneous) Markov chain:

$$X_{t+1} \sim P_t^{(\mathcal{A})}(X_t) \quad \text{and} \quad Y_{t+1} \sim P_t^{(\mathcal{B})}(Y_t), \quad (21)$$

where each $P_t^{(\mathcal{A})}$ or $P_t^{(\mathcal{B})}$ may depend on t . In many cases of multi-agent interactions, the *joint* chain (X_t, Y_t) is captured by:

$$(X_{t+1}, Y_{t+1}) \sim P_t(X_{t+1}, Y_{t+1} | X_t, Y_t), \quad (22)$$

where P_t again evolves over time (time inhomogeneity) to reflect policy changes, learning, or external interventions.

Encoding Cross-Agent Influences. The time-dependence in P_t can incorporate:

- Changes in *agent* $\mathcal{A}$ ’s internal motivations over time,
- Evolution in *agent* $\mathcal{B}$ ’s responsiveness to $\mathcal{A}$ ’s state,
- External manipulations or environmental shifts that alter how one agent influences the other.

Thus, each P_t can store the relevant *conditional probabilities* describing who influences whom (and how strongly) at each time step.

24.3 Sufficient Structure to Compute Time-Resolved Transfer Entropy

Obtaining $p_t(x_{t+1} | x_t, y_t)$. To calculate the TE in (20), one needs the conditional distribution $p_t(x_{t+1} | x_t, y_t)$, as well as $p_t(x_{t+1} | x_t)$. In the Markov context,

$$p_t(x_{t+1}, x_t, y_t) = p_t(y_t) p_t(x_t | y_t) p_t(x_{t+1} | x_t, y_t) \quad (\text{or an equivalent factorization}),$$

but more generally, we rely on the *joint* transition rule:

$$p_t(x_{t+1}, y_{t+1} | x_t, y_t).$$

Given that time-inhomogeneous Markov chains *encode* (via P_t) how (x_t, y_t) map to (x_{t+1}, y_{t+1}) , we can marginalize appropriately to find $p_t(x_{t+1} | x_t, y_t)$ and so forth. Thus, *all* the conditional probabilities needed for TE are derivable from the chain’s transitions.

Time Inhomogeneity and TE. Because P_t (the joint or pairwise transitions) depends on t , the TE $T_{Y \rightarrow X}(t)$ must also be considered on a *per-time-step basis*, often called *time-resolved* or *time-local* Transfer Entropy ?. Concretely, one interprets $p_t(\cdot)$ at each discrete t in (20). Hence, a *time-inhomogeneous* chain lets us see how $T_{Y \rightarrow X}(t)$ fluctuates or shifts as the agent(s) adapt or external conditions change.

24.4 Tracking Evolving Behavioral Influences Over Time

Once we can evaluate (or estimate)

$$T_{Y \rightarrow X}(t), \quad T_{X \rightarrow Y}(t) \quad \text{for each } t, \quad (23)$$

we obtain a *temporal profile* of how strongly agent Y influences agent X (and vice versa). A higher $T_{Y \rightarrow X}(t)$ indicates that X_{t+1} depends strongly on Y_t at time t , over and above X_t itself. Because the chain is *time inhomogeneous*, one can pinpoint *when* (or *if*) an intervention, new policy, or learning event causes the influence from Y to X to rise or fall.

Practical Example. Suppose agent $\mathcal{A}$ receives an external *prompt* at time t^* that modifies its free-energy-selection policy. The Markov transitions for $\mathcal{A}$ jump from $P_t^{(\text{old})}$ to $P_t^{(\text{new})}$ for $t \geq t^*$. Simultaneously, agent $\mathcal{B}$'s transitions may continue or also adjust if $\mathcal{B}$ depends on $\mathcal{A}$. Estimating time-resolved TE will show a discontinuity or marked change in $T_{\mathcal{B} \rightarrow \mathcal{A}}(t)$ or $T_{\mathcal{A} \rightarrow \mathcal{B}}(t)$ around t^* , *quantifying* the shift in causal influence.

24.5 Conclusion

Time-inhomogeneous Markov chains inherently encode enough conditional and joint probability information at each step t to support the estimation of (time-resolved) Transfer Entropy.

Because a Markov chain $P_t(x \rightarrow x')$ fully describes how an agent's behavioral tendencies evolve *given* prior states (and in multi-agent settings, given the other agent's states), one can recover:

1. The relevant conditional distributions $p_t(x_{t+1} \mid x_t, y_t)$,
2. The marginal $p_t(x_{t+1} \mid x_t)$,
3. Hence, the *directed influence* measure $T_{Y \rightarrow X}(t)$ as a function of time.

This is precisely the data needed to track *causal-like interactions* or *behavioral influences* among agents whose internal policies or environment can shift over time. As such, modeling multi-agent *free-energy deltas* or *behavioral states* with a time-inhomogeneous chain provides a sound, data-driven basis for computing Transfer Entropy in *nonstationary* or *adaptive* scenarios ??.

Key Takeaway: *Time-inhomogeneous Markov chains encode sufficient information about evolving behavioral tendencies to compute and interpret Transfer Entropy, thereby revealing how agents influence each other's dynamics in a directed manner over time.*

25 Background: Attachment Theory, Personality Traits, and Power Dynamics

Before developing our mathematical framework for modeling social-behavioral dynamics, we introduce three fundamental psychological theories that inform our approach: Attachment Theory, the Five-Factor Model of personality (commonly known as the Big Five), and Power Dynamics in social relationships.

25.1 Attachment Theory: Core Concepts

Attachment Theory, pioneered by Bowlby and expanded by Ainsworth, provides a framework for understanding how individuals form and maintain emotional bonds. The theory posits that early relationships with caregivers create internal working models that influence relationship patterns throughout life ?. These patterns manifest through three key behavioral systems:

- (i) **Proximity Maintenance:** The drive to establish and maintain closeness to attachment figures, particularly during stress or threat.
- (ii) **Safe Haven:** The tendency to return to an attachment figure for comfort and safety when faced with a perceived threat.
- (iii) **Secure Base:** The use of the attachment figure as a base of security from which to explore the environment.

These systems interact dynamically, creating observable patterns of interpersonal behavior. Individual differences in attachment manifest along two primary dimensions:

- **Attachment Anxiety:** The degree to which an individual fears abandonment and seeks excessive reassurance. Characterized by a negative internal model of self and a positive internal model of others.
- **Attachment Avoidance:** The extent to which an individual maintains emotional distance and self-reliance. Characterized by a negative internal model of others and a positive internal model of self.

The combination of these dimensions yields distinct attachment styles (secure, anxious-preoccupied, dismissive-avoidant, and fearful-avoidant), which predict patterns of interpersonal behavior, emotional regulation, and social network formation ?.

25.2 The Five-Factor Model of Personality

The Five-Factor Model (FFM) or "Big Five" represents the dominant paradigm in trait psychology, emerging from decades of factor-analytic studies ?. The model identifies five broad, independent dimensions of personality:

- (i) **Openness to Experience:** Reflects the breadth, depth, and complexity of an individual's mental life and experiences.
 - Characterized by curiosity, creativity, and willingness to engage with novel ideas
 - Manifests in behavioral flexibility and exploration of new patterns
- (ii) **Conscientiousness:** Describes the tendency to be organized, responsible, and hardworking.
 - Associated with goal-directed behavior and adherence to norms
 - Reflected in consistency and predictability of actions
- (iii) **Extraversion:** Captures the tendency to seek stimulation in the company of others.
 - Characterized by sociability, assertiveness, and energy
 - Visible in patterns of social network formation and influence
- (iv) **Agreeableness:** Represents the tendency to be compassionate and cooperative rather than suspicious towards others.
 - Manifests in prosocial behavior and relationship maintenance
 - Influences the formation of stable social bonds
- (v) **Neuroticism:** Reflects the tendency to experience negative emotions.
 - Associated with emotional instability and stress reactivity
 - Visible in the variability of behavioral patterns

25.3 Power Dynamics in Social Systems

Power dynamics are fundamental to understanding social relationships and behavioral influence. Following Dahl's seminal conceptualization, power represents "the probability that one actor within a social relationship will be in a position to carry out his own will despite resistance" ?. This framework encompasses several key components:

(i) **Asymmetric Interdependence:**

- Power manifests as unequal influence between actors in a social system
- Relationships typically involve bidirectional but asymmetric influence

(ii) **Resistance and Change:**

- Power involves overcoming others' resistance to change
- Effectiveness measured by ability to induce behavioral changes that wouldn't otherwise occur

(iii) **Institutional Power:**

- Emerges from persistent structures of aligned behaviors
- Creates stable patterns of influence across multiple actors

(iv) **Power Projection:**

- Ability to maintain influence even without direct interaction
- Often manifests through shared ideologies or established norms

Power relationships can be analyzed at multiple levels:

- **Individual:** Direct influence between pairs of actors
- **Group:** Collective power of coordinated actors
- **Institutional:** Emergent power from persistent social structures
- **Systemic:** Overall distribution and stability of power relations

25.4 Integration with Mathematical Modeling

These psychological frameworks provide natural targets for mathematical modeling, particularly through network analysis and stochastic processes:

- **Attachment patterns** can be conceptualized as dynamic network phenomena, where proximity seeking, safe haven behavior, and secure base provision manifest as measurable patterns of network formation and evolution.
- **Personality traits** can be understood as stable parameters that influence the probability distributions of behaviors and social interactions, making them amenable to analysis through Markov processes and network metrics.
- **Power relationships** can be quantified through asymmetric influence measures, resistance to behavioral change, and the persistence of induced changes over time, captured through transfer entropy and transition matrix stability.

In the following sections, a mathematical framework that captures these psychological constructs through quantifiable measures derived from dynamic causal graphs and time-inhomogeneous Markov chains will be presented. This approach allows us to connect qualitative psychological theory and precise mathematical modeling of social-behavioral dynamics.

26 Constructing a Dynamic Causal Graph via Pairwise Transfer Entropy

We now turn to the problem of constructing a *dynamic causal graph* or *directed behavioral-influence network* among multiple agents, each of which is modeled by a time-inhomogeneous Markov chain over free-energy deltas. Concretely, we describe how to compute *pairwise* Transfer Entropy (TE) in a time-resolved manner, then assemble these pairwise TE values into a *dynamic* or *time-indexed* adjacency matrix that reflects the causal-like influence each agent exerts on the others.

26.1 Preliminaries: Multiple Agents and Free-Energy Deltas

Assume we have M agents, labeled $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_M$. For each agent $\mathcal{A}_i$, we observe a time-indexed process

$$\{X_t^{(i)}\}_{t=0}^T,$$

where $X_t^{(i)}$ denotes the *free-energy delta* (or a relevant discretized state representing ΔF) at time t . As argued in previous sections, each $\{X_t^{(i)}\}$ can be viewed as a *time-inhomogeneous* Markov chain whose transitions incorporate the agent’s evolving motivations, policies, or external influences.

Joint Observation Across Agents. At each discrete step t , we thus have a state vector

$$\mathbf{X}_t = (X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(M)}).$$

While each chain $X_t^{(i)}$ can be estimated separately (by $\{P_t^{(i)}\}$), the *joint* transitions of $\mathbf{X}_t$ may encode cross-agent dependencies in a time-varying fashion, capturing the possibility that agent $\mathcal{A}_i$ ’s next state depends not only on $\mathcal{A}_i$ ’s current state but also on others’ states.

26.2 Pairwise Transfer Entropy for Causal Inference

Definition of Time-Resolved TE. Given two agents, $\mathcal{A}_i$ and $\mathcal{A}_j$, one computes a time-local or time-resolved Transfer Entropy ??:

$$T_{\mathcal{A}_j \rightarrow \mathcal{A}_i}(t) = \sum_{x_{t+1}^{(i)}, x_t^{(i)}, x_t^{(j)}} p_t(x_{t+1}^{(i)}, x_t^{(i)}, x_t^{(j)}) \ln \left[\frac{p_t(x_{t+1}^{(i)} | x_t^{(i)}, x_t^{(j)})}{p_t(x_{t+1}^{(i)} | x_t^{(i)})} \right]. \quad (24)$$

Here, $p_t(\cdot)$ denotes the (possibly time-varying) joint or conditional probabilities at time t . Intuitively, $T_{\mathcal{A}_j \rightarrow \mathcal{A}_i}(t)$ measures how agent $\mathcal{A}_j$ ’s state at time t helps predict $\mathcal{A}_i$ ’s state at time $t+1$, beyond what $\mathcal{A}_i$ ’s own prior state reveals.

Relevance to Causality. A higher $T_{\mathcal{A}_j \rightarrow \mathcal{A}_i}(t)$ indicates a stronger directed (or “causal-like”) influence exerted by $\mathcal{A}_j$ on $\mathcal{A}_i$ at step t . While TE is formally an *information-theoretic* measure of predictive power, it is often interpreted—under Markov assumptions and appropriate controls for confounders—as indicating causal or “who drives whom” relationships ?.

26.3 Constructing a Dynamic Causal Graph

To build a *dynamic causal (behavioral-influence) graph* among the M agents, one proceeds as follows:

1. **Estimate the time-inhomogeneous Markov chain for each agent’s ΔF :**

From empirical data (or simulation), fit $\{P_t^{(i)}\}$ for each agent $\mathcal{A}_i$, capturing how $X_{t+1}^{(i)}$ depends on $X_t^{(i)}$ and possibly other agents’ states $X_t^{(j)}$, $j \neq i$. If necessary, one may model the joint transitions $P_t(\mathbf{X}_t \rightarrow \mathbf{X}_{t+1})$ directly.

2. Compute TE pairwise at each time step:

For each pair $(\mathcal{A}_i, \mathcal{A}_j)$ and each $t = 0, 1, \dots, T-1$, calculate $T_{\mathcal{A}_j \rightarrow \mathcal{A}_i}(t)$ using (24). This requires the conditional probabilities

$$p_t(x_{t+1}^{(i)} | x_t^{(i)}, x_t^{(j)}) \quad \text{and} \quad p_t(x_{t+1}^{(i)} | x_t^{(i)}),$$

both of which are derivable or estimable from the time-inhomogeneous Markov structure.

3. Form a time-indexed adjacency matrix:

For each time step t , create an $M \times M$ matrix $\mathbf{W}(t)$, where the directed edge weight from agent j to agent i is

$$W_{i,j}(t) = T_{\mathcal{A}_j \rightarrow \mathcal{A}_i}(t).$$

This yields a *directed* graph (or adjacency matrix) at each t . If desired, one can threshold $T_{\mathcal{A}_j \rightarrow \mathcal{A}_i}(t)$ to form a binary edge indicator (i.e., whether the influence is “significant”).

4. Obtain a dynamic or time-varying graph:

By stacking or indexing the matrices $\{\mathbf{W}(t)\}_{t=0}^{T-1}$, we get a *dynamic causal graph* that evolves with t . One can visualize how an edge from agent j to agent i becomes stronger or weaker over time, reflecting changes in how agent j ’s free-energy deltas influence agent i ’s subsequent deltas.

Interpretation. The resulting dynamic graph can be read as a *behavioral-influence network*: at each time step, TE reveals which agent’s free-energy changes most effectively (or predictively) drive the other’s next state. Because the underlying Markov chains are *time inhomogeneous*, one directly captures adaptation or external interventions that alter the coupling strengths among agents.

Illustrative Example

If at time t_* agent $\mathcal{A}_k$ experiences a policy shift that makes it more sensitive to $\mathcal{A}_\ell$ ’s actions, we expect $T_{\mathcal{A}_\ell \rightarrow \mathcal{A}_k}(t)$ to *increase* for $t \geq t_*$. Simultaneously, if $\mathcal{A}_k$ becomes less influential on $\mathcal{A}_m$, then $T_{\mathcal{A}_k \rightarrow \mathcal{A}_m}(t)$ might *decrease*. Plotting $\mathbf{W}(t)$ over time exposes these shifts in the directed edge weights.

26.4 Why Markov Chains on ΔF Are Sufficient for TE

As argued in Section 24, a time-inhomogeneous Markov chain can store all needed conditional probabilities $p_t(x_{t+1}^{(i)} | x_t^{(i)}, x_t^{(j)})$. Since Transfer Entropy is essentially a ratio of such conditional probabilities (see Eq. (24)), we affirm that *no additional or hidden variables* are strictly required, provided that the chain’s state space *already* encodes the relevant influences. Hence, a well-fitted chain for each agent’s free-energy deltas is sufficient to compute TE and build a dynamic causal graph of multi-agent behavioral influences.

26.5 Conclusion

We have shown how to construct a *dynamic causal graph* among multiple agents by:

- (a) Modeling each agent’s free-energy deltas with a *time-inhomogeneous Markov chain*,
- (b) Computing *pairwise* Transfer Entropy at each time step,
- (c) Assembling these pairwise TE values into a *time-resolved* adjacency matrix.

The resulting network reveals, at each time t , the directed edge weights $T_{\mathcal{A}_j \rightarrow \mathcal{A}_i}(t)$ —i.e., how strongly agent $\mathcal{A}_j$ ’s free-energy changes *predict* or *drive* agent $\mathcal{A}_i$ ’s next changes. Thus, we obtain a *behavioral influence network* that evolves in tandem with the agent’s (or agents’) policy changes, external influences, or nonstationary environment.

27 Connecting the Dynamic Causal Graph to Personality and Attachment Metrics

Having established a dynamic causal network among agents (Section 26), we now describe how this *evolving influence graph*—alongside each agent’s underlying time-inhomogeneous Markov chain of free-energy deltas—permits quantitative inferences about:

- (a) **Attachment dynamics** (e.g., proximity seeking, safe haven, secure base),
- (b) **Personality traits** under the Big Five model (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism).

We will see that classical **network measures** (e.g., degree, clustering, betweenness) and **Markov chain variability metrics** (e.g., transition entropy, consistency) serve as robust proxies for these behavioral dispositions.

27.1 Attachment Dynamics via Evolving Network Measures

From an *attachment theory* standpoint ??, individuals exhibit:

- **Proximity seeking** (desire for closeness),
- **Safe haven** (turning to certain figures or networks under stress),
- **Secure base** (the individual’s role in supporting or anchoring others),
- **Attachment stability** (consistency or variability over time).

We propose to interpret certain *network centrality* and *connectivity* measures—computed on the dynamic causal graph at each time step—as approximate indicators of attachment behaviors.

27.1.1 Degree Centrality (Proximity Seeking)

- **High degree centrality:** An agent has many directed edges (incoming or outgoing) in the causal graph.
 - Implies the agent frequently interacts or is influenced by multiple others (incoming edges) or exerts influence on them (outgoing edges).
 - Interpreted as an agent *seeking proximity*, i.e., engaging with or attracting many contacts.
- **Low degree centrality:**
 - The agent maintains fewer or weaker direct connections over time.
 - Could reflect less interest in social closeness or an avoidance of strong ties.

27.1.2 Clustering Coefficient (Safe Haven)

- **High clustering coefficient:**
 - In a local neighborhood, the agent’s contacts are heavily interconnected.
 - Suggests membership in a *tight-knit group*, offering emotional or instrumental support.
 - Reflects a robust “*safe haven*” network where the agent can reliably find help or refuge.
- **Low clustering coefficient:**
 - The agent’s neighbors do not strongly connect among themselves.
 - May indicate social fragmentation—lack of a cohesive, supportive cluster.
 - Potentially signifies isolation or an environment of loosely connected, uncoordinated relationships.

27.1.3 Betweenness Centrality (Secure Base)

- **High betweenness centrality:**

- The agent frequently lies on shortest paths between other pairs of nodes.
- Functions as a *bridge* or *hub*, facilitating information flow or emotional support.
- Interpreted as a “secure base,” given that others may *rely on* this agent for connections or resources.

- **Low betweenness centrality:**

- The agent rarely acts as a connector between distinct subgroups.
- Suggests a more peripheral or self-contained role, not serving as others’ anchor.

27.1.4 Variance in Network Metrics (Attachment Stability)

- **Low variance over time:**

- The agent’s centralities, clustering, or other network measures remain stable across the dynamic graph snapshots.
- Implies *consistent attachment behavior*—predictable, stable patterns of seeking, safe haven, or secure base roles.

- **High variance:**

- The agent’s position or connectivity fluctuates markedly over different time intervals.
- Reflects *instability* or inconsistency in attachment-related behavior, possibly due to crises, external stressors, or internal conflict.

Longitudinal Attachment Pattern Recognition. Because our causal graph updates each time step t , we can trace *how* an agent’s attachment-like indices (degree, betweenness, clustering) evolve over time. This dynamic approach reveals how attachment styles may shift in response to new relationships, major life events, or therapeutic interventions ?.

27.2 Big Five Personality Dimensions from the Markov Transition and Network Dynamics

Many trait-psychology theories (e.g., the Big Five model) emphasize consistent behavioral tendencies *and* the capacity for adaptation in different social contexts ?. Using the agent’s time-inhomogeneous Markov chain(s) and the evolving network structure, we approximate each of the five traits as follows:

27.2.1 Openness (Behavioral Flexibility)

- **Metric: Behavioral Flexibility.**

- Evaluate changes in each agent’s transition matrix $\{P_t^{(i)}\}$ across time steps.
- Track introduction of *new* transitions or increased *entropy* in transitions, indicating exploratory or novel behavior.

- **Interpretation.**

- High *Openness*: Agent exhibits wide behavioral repertoire, readiness to try new behaviors (non-zero transitions not seen before).
- Low *Openness*: Agent’s transitions remain narrow or repetitive, minimal shift in entropy or diversity of actions.

27.2.2 Conscientiousness (Behavioral Consistency)

- **Metric: Consistency Over Time.**

- Compare consecutive transition matrices $\{P_t^{(i)}\}$ to see how stable the agent’s action probabilities are.
- E.g., measure matrix similarity (e.g., via Frobenius norm or Kullback–Leibler divergence).

- **Interpretation.**

- High *Conscientiousness*: Agent shows reliable, organized patterns, with minimal abrupt changes in transitions.
- Low *Conscientiousness*: Marked fluctuations or irregularities in transitions, reflecting impulsivity or disorganization.

27.2.3 Extraversion (Network Centrality)

- **Metric: Social (Causal) Centrality.**

- Use network centrality measures in each snapshot: degree, betweenness, eigenvector centrality, etc.
- Agents with consistently high centrality across time are highly “extraverted” in the sense of being socially active and influential.

- **Interpretation.**

- High *Extraversion*: Frequent or robust causal connections with many others, acting as a key influencer or bridging node.
- Low *Extraversion*: Smaller or fewer direct connections, more peripheral role in the network.

27.2.4 Agreeableness (Cooperative Network Behavior)

- **Metric: Cooperative and Reciprocal Interactions.**

- Clustering coefficient: Tendency to form *cohesive*, mutually supportive clusters.
- Reciprocity ratio: Fraction of edges $(A \rightarrow B)$ that have a counterpart $(B \rightarrow A)$ with non-trivial weight.
- Stability of these mutual edges over time.

- **Interpretation.**

- High *Agreeableness*: Maintains stable, cooperative ties, forming highly reciprocal or cluster-like structures in the dynamic network.
- Low *Agreeableness*: Tends toward antagonistic or uncooperative behaviors, with fewer mutual or stable connections.

27.2.5 Neuroticism (Behavioral Volatility)

- **Metric: Volatility of Transition Patterns.**

- Track *variance* or *high rates of change* between $\{P_t^{(i)}\}$ from one timestep to the next.
- Possibly measure standard deviation of transition probabilities or “change rate” across time windows.

- **Interpretation.**

- High *Neuroticism*: Rapid or erratic shifts in behavioral transitions, indicating emotional reactivity or instability.
- Low *Neuroticism*: Calm, stable transition patterns, less prone to sudden changes.

27.3 Putting it All Together: A Comprehensive Behavioral Model

Multiple Lenses on the Same Data. Both *attachment dynamics* (Section 27.1) and *Big Five traits* rely on:

- (i) The time-resolved *causal graph*—to derive network measures (degree, clustering, betweenness, reciprocity, etc.).
- (ii) The *time-inhomogeneous Markov chain*—to compute transitions’ entropy, stability, or diversity (key for Openness, Conscientiousness, Neuroticism).

This synergy ensures that we capture both an agent’s *internal consistency* (via Markov transitions) and *social role* (via network centralities).

Longitudinal Tracking of Personality and Attachment. Because the system updates each step in a discrete timeline $t = 0, 1, \dots, T$, we can track how these metrics evolve—yielding insights into *when* an agent’s openness spikes, or how betweenness centrality changes following a stressful event. This approach naturally supports *longitudinal* or *narrative* studies where personality traits and attachment patterns shift over key episodes.

Integration for Applied Settings. Finally, these methods can be used in:

- **Group therapy or family systems:** Identifying which members function as “secure base” or bridging connectors, how individuals’ volatility affects group stability.
- **Clinical intervention:** Monitoring attachment dynamics over therapeutic sessions, detecting heightened “proximity seeking” or new “avoidant” patterns from network changes.
- **Cross-cultural or comparative research:** Quantitatively comparing how strongly certain Big Five traits or attachment roles manifest in different populations by analyzing the distribution of centrality, transition diversity, or variance metrics.

The dynamic-causal-graph plus Markov-based personality measurement thus provides a robust computational lens for real-world data on social and behavioral processes.

28 Extensions of Matrix Transfer Entropy to Indirect, Common, and Synergistic Causation

Having introduced Matrix Transfer Entropy (MTE) as a robust and computationally efficient alternative to classical Transfer Entropy, we now discuss how MTE can be extended or interpreted to capture more nuanced causal structures among multiple variables. In particular, we show how one can use MTE to (i) detect indirect or mediating pathways, (ii) disentangle common (or confounding) causal drivers, and (iii) identify synergistic effects where joint drivers exert more influence than each alone. Below, we outline the conceptual and computational rationale for each.

28.1 Indirect (Mediated) Causation

Conceptual Setup. In many real-world scenarios, a variable X may influence Z *only* through an intermediate variable Y , a phenomenon known as *indirect* or *mediated* causation:

$$X \longrightarrow Y \longrightarrow Z, \quad (\text{but no direct pathway } X \rightarrow Z).$$

Standard pairwise TE (or MTE) between X and Z might show a small direct effect, whereas the chain $X \rightarrow Y$ and $Y \rightarrow Z$ is strong.

Operationalizing Indirect Causation with MTE. To capture mediation in an MTE context:

- (i) **Compute pairwise MTE** among $(X \rightarrow Y)$, $(Y \rightarrow Z)$, and $(X \rightarrow Z)$.
- (ii) **Compare** the direct $MTE_{X \rightarrow Z}$ against the product or composition of $MTE_{X \rightarrow Y}$ and $MTE_{Y \rightarrow Z}$.
- (iii) **Interpret:**
 - If $MTE_{X \rightarrow Z}$ is notably smaller than $MTE_{X \rightarrow Y}$ and $MTE_{Y \rightarrow Z}$, that indicates X influences Z *primarily* via Y .
 - If $MTE_{X \rightarrow Z}$ remains large even after partialing out Y (see next subsection), then X also has a *direct* path to Z , not purely indirect.

Because MTE is effectively a *model-free* measure of conditional dependence among the time-lagged variables, we can implement a partial or conditional MTE analysis (analogous to partial TE ?) to more rigorously test whether $X \rightarrow Z$ is mediated by Y .

28.2 Common (Confounding) Causation

Common Driver Scenario. Another frequently encountered pattern is *common causation*, in which a latent or external source W drives both X and Y :

$$W \longrightarrow X \quad \text{and} \quad W \longrightarrow Y, \quad (\text{but } X \text{ and } Y \text{ do not directly influence each other}).$$

Pairwise TE (or MTE) might erroneously suggest $X \leftrightarrow Y$ causation if W is unobserved. Even if W is partially observed, naive TE could conflate “common driver” influence with direct $X \rightarrow Y$ or $Y \rightarrow X$.

Matrix Transfer Entropy Approach. To disentangle such confounding:

- (i) **Include** the putative common driver W in the MTE analysis if data are available.
- (ii) **Compute partial or conditional MTE**, for instance $MTE_{X \rightarrow Y|W}$, which measures how much X adds to predicting Y once W is accounted for.
- (iii) **Compare** $MTE_{X \rightarrow Y|W}$ against the unconditional $MTE_{X \rightarrow Y}$.
 - If $MTE_{X \rightarrow Y|W}$ drops (near zero) while $MTE_{X \rightarrow Y}$ was large, that indicates W is the *true* source of Y (common cause).
 - If $MTE_{X \rightarrow Y|W}$ remains sizable, then X indeed has a direct effect on Y beyond the confounding effect of W .

Because MTE is typically lower in computational cost than classical TE (due to bypassing PDF estimation), partial MTE in a multi-variate setting can be more tractable.

28.3 Synergistic Effects

Definition of Synergy. Synergy arises when two (or more) driver variables X_1, X_2 *jointly* influence Y in a way that exceeds the sum of their separate influences. Formally, synergy can be tested by comparing:

$$MTE_{(X_1, X_2) \rightarrow Y} \quad \text{versus} \quad MTE_{X_1 \rightarrow Y} + MTE_{X_2 \rightarrow Y}$$

or by using an established synergy or “unique information” decomposition ? in an MTE context.

Implementing Synergy with MTE. One can define a multi-variable MTE measure, e.g. $MTE_{(X_1, X_2) \rightarrow Y}$, where (X_1, X_2) is taken as a joint predictor. If

$$MTE_{(X_1, X_2) \rightarrow Y} > MTE_{X_1 \rightarrow Y} + MTE_{X_2 \rightarrow Y},$$

this indicates that *together*, X_1 and X_2 contain more predictive power about Y than they do individually. That is the hallmark of a *synergistic* effect: combining the drivers yields new predictive information not found in either alone.

Practical Computation. To compute $MTE_{(X_1, X_2) \rightarrow Y}$, one constructs the relevant Gram matrices for the joint lagged states (X_1, X_2) and uses the same matrix-entropy approach as in the pairwise MTE formula (Section 3 of your MTE framework). This is generally more compute-efficient than naive multi-dimensional PDF estimation, particularly if X_1 and X_2 have many samples or strong nonlinearities.

28.4 Summary of Multi-Variable MTE Benefits

- **Reduced Complexity:** Since Matrix Entropy bypasses explicit probability density estimations, partial or multi-variable MTE is significantly more tractable than typical multi-variate TE.
- **Robustness to Noise:** Like the pairwise MTE, multi-agent or partial MTE remains less sensitive to heavy-tailed or non-Gaussian noise.
- **Decomposability:** By systematically computing $MTE_{X \rightarrow Y}$, $MTE_{X \rightarrow Y|Z}$, or $MTE_{(X_1, X_2) \rightarrow Y}$, one can identify indirect paths ($X \rightarrow Y$ through Z), common cause confusions, and synergy among multiple drivers.

Hence, these extensions preserve the computational efficiency and noise robustness advantages of MTE while enabling more sophisticated *causal structure* inferences.

28.5 Implementation Notes

Data Requirements. As with classical TE, reliable partial or multi-driver MTE requires enough data to estimate the relevant Gram matrices for the expanded variable sets (X, Y, Z) , etc. If the dimension grows large, one might need sufficient sample size to maintain stable matrix-based estimates. However, in many practical scenarios—especially compared to direct PDF-based TE—the sample-size requirements for MTE remain more moderate.

Practical Steps.

- Identify* the potential set of drivers or confounders, e.g. W in a common-cause scenario, or (X_1, X_2) in synergy.
- Construct* the appropriate Gram matrices for the (lagged) joint states.
- Compute* the MTE measures of interest (conditional MTE, multi-driver MTE).
- Compare* unconditional and conditional values to reveal indirect, confounded, or synergistic influences.

Link to the Larger System. In a multi-agent scenario where each agent’s free-energy deltas are time-series $\Delta F^{(i)}$, one might:

- Use pairwise MTE for initial causal graphs (Sec. 26).
- Then refine or dissect certain edges by partial or synergy-based MTE, clarifying whether an edge $X \rightarrow Y$ is direct or indirect, purely confounded, or strongly synergistic with a third agent Z .

Conclusion of Extensions

Matrix Transfer Entropy, originally introduced as a more efficient and robust method for two-variable causality detection, naturally generalizes to multi-variate settings. By computing partial MTE or multi-driver MTE, one can detect:

- Indirect (mediated) causation*, by testing $X \rightarrow Z$ once the mediator Y is included;
- Common-driver* or *confounding* effects, via conditional MTE $X \rightarrow Y | W$;
- Synergy* among multiple drivers (X_1, X_2) with respect to a target Y .

All of these analyses preserve MTE’s strengths, namely *avoiding explicit PDF estimation* and *maintaining noise robustness*. As such, MTE can serve not only as a simpler pairwise measure but also as a comprehensive tool for unraveling complex causal structures in multi-agent or multi-variable systems.

29 Modeling Environmental Influences and the Markov Blanket via Time-Inhomogeneous Markov Chains of Free-Energy Deltas

We have established that each agent’s behavioral tendencies can be tracked through a time-inhomogeneous Markov chain whose state space represents *free-energy deltas* (ΔF), capturing how much free energy is locally increased or decreased with each action or step. This section shows how we incorporate *environmental influences* in such a Markov setting and relate them to the *Markov Blanket* framework from Active Inference.

29.1 Environmental Influences in a ΔF -Based Markov Chain

Agent Free-Energy Deltas. Recall that for each agent $\mathcal{A}_i$, we have a Markov chain in discrete time steps $t = 0, 1, \dots$, whose state is $\Delta F_t^{(i)}$. This $\Delta F_t^{(i)}$ is a (possibly discretized) variable describing how the agent’s free energy changes from step t to $t + 1$. Because the agent’s ΔF transitions can depend on the agent’s own *internal factors* (preferences, drives) and on *external* or *environmental* conditions, we model the transition probabilities as *time-inhomogeneous*:

$$P_t(\Delta F_{t+1}^{(i)} \mid \Delta F_t^{(i)}) \text{ varies with } t.$$

Hence, each step’s update rule can adapt over time, reflecting that both the agent’s internal states and the external environment can shift policies or constraints.

Incorporating Environment States. If we wish to embed *environmental influences* explicitly, we can treat environment-related variables as either:

1. **Observed** external signals that affect $\Delta F_{t+1}^{(i)}$, or
2. **Hidden** states whose fluctuations shape the transitions for $\Delta F_{t+1}^{(i)}$ but are not directly observed.

In the first scenario, we might define a joint chain $\{(\Delta F_t^{(i)}, E_t)\}$, where E_t denotes environment states or signals at time t . If the environment is unobserved, we can define a hidden-state extension (akin to a Hidden Markov Model) so that each agent’s $\Delta F_{t+1}^{(i)}$ is conditionally dependent on E_t . In either case, the environment’s influences become part of the transition kernel

$$P_t(\Delta F_{t+1}^{(i)} \mid \Delta F_t^{(i)}, E_t).$$

This encodes how changes in free energy can be *driven* or *modulated* by environment conditions.

Time Inhomogeneity via Environmental Dynamics. Time inhomogeneity emerges naturally if the environment itself changes over time in a nonstationary way (e.g., new constraints, seasons, social contexts). If we track $\{E_t\}$ as a (hidden) nonstationary process, then the agent’s ΔF -chain inherits time-dependent transitions.

29.2 Connecting to the Markov-Blanket Notion (Active Inference View)

Markov Blanket in a Discrete Setting. Active Inference typically treats an agent’s states as partitioned into *internal*, *external*, and *blanket* subsets:

- External $\leftrightarrow$ The environment,
- Internal $\leftrightarrow$ The agent’s hidden (interoceptive or neuronal) states,
- Blanket $\leftrightarrow$ Sensory and active states that separate internal from external.

In a *discrete-time, ΔF -based model*, we can approximate this by saying:

$$Z_t^{(i)} = (I_t^{(i)}, B_t^{(i)}, E_t),$$

where:

- $I_t^{(i)}$: Agent $\mathcal{A}_i$'s *internal* hidden variables (could be the portion of ΔF that is “private” to the agent’s internal processes),
- $B_t^{(i)}$: Blanket states relevant to agent $\mathcal{A}_i$ (e.g., control actions, sensors),
- E_t : The environment states (or environment’s influence on ΔF).

Since $\Delta F_{t+1}^{(i)}$ can be viewed as a specific function of $\Delta F_t^{(i)}$ plus environment ΔF interactions, these sub-components can reflect how ΔF is “partitioned” into internal vs. environment-driven changes.

Markov-Blanket Factorization. A hallmark property in Active Inference is that the external and internal states only influence each other “through” the blanket states. Symbolically, if we define transitions in a time-inhomogeneous setting, we want:

$$p_t\left(I_{t+1}^{(i)}, B_{t+1}^{(i)}, E_{t+1} \mid I_t^{(i)}, B_t^{(i)}, E_t\right) = p_t(I_{t+1}^{(i)} \mid I_t^{(i)}, B_t^{(i)}) \cdot p_t(B_{t+1}^{(i)} \mid I_{t+1}^{(i)}, E_t) \cdot p_t(E_{t+1} \mid E_t, B_t^{(i)}),$$

as one possible discrete Markov-blanket factorization. In simpler terms, ΔF changes from internal and external influences, but *internal* cannot “see” external states except via blanket states, and vice versa.

Why This Matters. It shows “where” environment states enter the agent’s ΔF transitions. The blanket states channel how the environment modifies $\Delta F_t^{(i)}$ or how the agent acts on the environment. This is entirely consistent with the notion that in an Active-Inference-style system, the environment sets constraints or opportunities, shaping free-energy dynamics, and the agent’s internal states (plus blanket) produce outward actions or sense data.

29.3 Transfer Entropy (TE) Among Agent’s ΔF States and Environment

Hidden or Observed Environment. Once we embed the environment as $\{E_t\}$ (hidden or observed), we can measure *Transfer Entropy* among:

$$\Delta F^{(i)} \quad \text{and} \quad E_t,$$

to see how changes in environment states drive the agent’s free-energy deltas, or conversely, how strongly the agent’s ΔF influences environment states if the agent actively modifies them. In a multi-agent scenario, we can add environment states to the matrix Transfer Entropy approach and build a bigger directed causal graph.

Matrix TE or Partial TE. - If we have multiple agents $\{\mathcal{A}_i\}$ plus environment E , one might do “Matrix TE” pairwise or partial TE to factor out confounders. - For instance, we can compute:

$$T_{E \rightarrow \Delta F^{(i)}} \quad \text{and} \quad T_{\Delta F^{(i)} \rightarrow E} \quad \text{for each } i,$$

giving us a sense of “who influences whom” with respect to environment $\leftrightarrow$ agent couplings. The *time inhomogeneity* means these conditional probabilities vary at each step t , capturing a dynamic environment or agent.

Interpretation. High $T_{E \rightarrow \Delta F^{(i)}}$ means the environment states significantly drive or predict the agent’s free-energy deltas. High $T_{\Delta F^{(i)} \rightarrow E}$ means the agent’s behaviors are actively altering environment conditions. A Markov-blanket factorization suggests that if we partition states properly, environment $\leftrightarrow$ internal interactions remain indirect through blanket variables (which might be part of the agent’s ΔF or partially observed).

29.4 Summary and Key Takeaways

1. Environmental Embedding in ΔF -Chains. By extending the time-inhomogeneous Markov chain for each agent’s ΔF to include environment states E_t (hidden or observed), we can:

- Capture how external conditions shape $\Delta F_{t+1}^{(i)}$,
- Reflect nonstationary or adaptive environment changes over t ,
- Potentially separate environment-driven deltas from internal-driven deltas.

2. Connection to Markov Blankets. A discrete Markov-blanket factorization can be placed on $(I_t^{(i)}, B_t^{(i)}, E_t)$ so that internal states of agent i and external environment only interact through a “blanket.” This is consistent with Active Inference’s core structure, recast in a time-inhomogeneous chain for free-energy deltas.

3. TE-based Causal Graph. Finally, measuring *Transfer Entropy* among these states (or their posteriors, if hidden) yields a *dynamic causal graph*:

- Edges from environment E_t to $\Delta F_{t+1}^{(i)}$ (i.e. $T_{E \rightarrow \Delta F^{(i)}}$) show environment’s direct influence on the agent’s free-energy changes.
- Edges from $\Delta F_t^{(i)}$ to environment E_{t+1} reveal how the agent modifies environmental conditions (e.g., actions).

When multiple agents are present, partial or matrix TE clarifies cross-agent plus environment couplings in a single network representation.

Concluding Remarks for This Section

By embedding the *environment* into a time-inhomogeneous Markov or hidden-Markov chain for ΔF -based states, and adopting a Markov-blanket decomposition if desired, we produce a powerful framework wherein:

1. ΔF retains its role as the agent’s local measure of free-energy changes (motivational or policy-driven),
2. External influences are modeled either as explicit observed states or hidden environment variables,
3. The Markov-blanket condition from Active Inference is realized in discrete-time transitions,
4. Transfer Entropy among these variables gives a *causal/influence graph* of how environment states shape each agent’s ΔF , and possibly how the agent modifies the environment in turn.

This lays out a coherent approach for bridging ΔF -centric time-inhomogeneous Markov modeling, environmental influences, and Active Inference’s Markov-blanket logic into one integrated causal analysis framework.

Below is a **self-contained theoretical proof** (in sketch form but with rigorous definitions) that *causal invariance* in Wolfram’s multiway rewriting system implies *maximal* (or near-maximal) **Transfer Entropy** (TE) among “observer” processes whose vantage branches *unify*. While this is still *somewhat schematic* (as a fully rigorous version would occupy many pages of definitions and set-up), it provides the key ingredients and logical steps to transform Wolfram’s notion of **causal invariance** into a strong statement about **transfer entropy** among multi-agent vantage processes.

Disclaimer:

1. We assume some familiarity with (i) Wolfram’s multiway rewriting systems, (ii) standard definitions of Transfer Entropy, and (iii) time-inhomogeneous Markov processes.
2. The structure below is flexible enough that details can be adapted to specific rewriting or concurrency formalisms (hypergraphs, string rewrites, etc.).

30 Mathematical Setup

30.1 Multiway Rewriting System

Definition 1 (Multiway System).

A *multiway rewriting system* is a tuple

$$\mathcal{W} = (\mathcal{H}, \rho, h_0),$$

where:

1. $\mathcal{H}$ is a set of *states* (e.g., hypergraphs, strings, or combinatorial objects).
2. $\rho : \mathcal{H} \rightarrow \mathcal{P}(\mathcal{H})$ is a *rewrite function*, mapping each state $h \in \mathcal{H}$ to the set of all possible “successor states.”
3. $h_0 \in \mathcal{H}$ is an *initial state*.

Multiway evolution: One obtains a directed acyclic graph (DAG) $M(\mathcal{W})$ whose nodes are the states reachable from h_0 . An edge (h_n, h_{n+1}) exists if $h_{n+1} \in \rho(h_n)$. Because we allow *all* rewrites to occur in parallel or in branching, we get multiple successors from each node, leading to a *branching* structure.

30.2 Events, Causal Hypergraph, and Causal Invariance

Definition 2 (Rewrite Events).

An *event* E is a labeled rewrite instance: “Applying rule α to sub-part $A \subseteq h_n$ to yield part $B \subseteq h_{n+1}$.” More formally, each edge $(h_n \rightarrow h_{n+1})$ in the multiway graph might be decomposed into multiple *events* if multiple rewrites happen in “parallel” or in separate sub-parts. Each event E transforms a portion of h_n .

Definition 3 (Causal Dependencies).

If an event E_1 modifies a region $\Delta \subset h_n$ upon which event E_2 also depends (i.e., E_2 attempts to rewrite or read the same region Δ), then we say $E_1 \prec E_2$ in the *causal ordering*.

Definition 4 (Causal Hypergraph).

Let $\mathbf{E}$ be the set of all events across the multiway evolution. Construct a directed hypergraph Γ whose nodes are events in $\mathbf{E}$. There is a directed link $E_1 \rightarrow E_2$ if $E_1 \prec E_2$. The partial order $\prec$ is typically *acyclic*.

Definition 5 (Causal Invariance).

We say $\mathcal{W}$ is *causally invariant* if, for *every* pair of distinct rewriting paths Π_1, Π_2 from the initial state h_0 , the corresponding sets of events $\mathbf{E}_1, \mathbf{E}_2$ become *isomorphic* in partial order after some finite stage. More concretely:

1. Possibly up to a relabeling or “isomorphism,” the partial order of events in Π_1 matches that in Π_2 .
2. No difference in the order in which events happen leads to a contradictory or irreconcilable final event net.

(Precise statements can be found in Wolfram’s expositions, but this is the essential property.)

31 Observers / Agents as Partial Vantages (Local Branches)

31.1 Agent Observations

Let there be N “agents” $\{\mathcal{A}_i\}_{i=1}^N$. Each agent sees or “subscribes” to a certain portion of the rewriting events. For instance, agent $\mathcal{A}_i$ may only track events that affect a region $\mathcal{R}_i \subseteq h$. As the multiway rewriting unfolds, each agent collects a *time series*:

$$X^{(i)} = \{X_0^{(i)}, X_1^{(i)}, X_2^{(i)}, \dots\},$$

where $X_t^{(i)}$ is the agent’s local “state” at time t .

Key: We define the notion of “time” for $\mathcal{A}_i$ so that each *rewrite event* that modifies $\mathcal{R}_i$ increments the local clock $t \mapsto t + 1$. This yields a *branch* in the multiway system that the agent “travels.” Possibly, many global rewriting steps occur in parallel each “agent-time” if the agent lumps them together, etc.

Hence each agent $\mathcal{A}_i$ effectively follows a *branch* Π_i . (Some branches might partially overlap or merge.)

31.2 Time-Inhomogeneous (Hidden) Markov Chains

One can interpret the local evolution $\{X_t^{(i)}\}$ as a time-inhomogeneous Markov process or a Hidden Markov chain:

1. **Time Inhomogeneity:** Because the set of events that can occur at step t may vary depending on prior rewriting, the transition probabilities

$$p_t(X_{t+1}^{(i)} | X_t^{(i)})$$

can change over time or with global branching.

2. **Hidden Markov:** If $\mathcal{A}_i$ only sees a partial or incomplete observation of the hypergraph, then we might define hidden states $\{Z_t^{(i)}\}$ that reflect the “true” local hypergraph region. The agent’s observed states $X_t^{(i)}$ are drawn from an emission model.

32 Transfer Entropy and Multi-Agent Causal Graph

Given two agents $\mathcal{A}_i$ and $\mathcal{A}_j$, define the *time-resolved* Transfer Entropy at each step t :

$$T_{X^{(i)} \rightarrow X^{(j)}}(t) = \sum_{x_{t+1}^{(j)}, x_t^{(j)}, x_t^{(i)}} p_t(x_{t+1}^{(j)}, x_t^{(j)}, x_t^{(i)}) \ln \left[\frac{p_t(x_{t+1}^{(j)} | x_t^{(j)}, x_t^{(i)})}{p_t(x_{t+1}^{(j)} | x_t^{(j)})} \right].$$

(One can also use Matrix Transfer Entropy for efficiency or partial TE for multi-driver scenarios. We keep it simple.)

By computing $T_{X^{(i)} \rightarrow X^{(j)}}(t)$ for all pairs (i, j) , we obtain a *directed network* or adjacency matrix that we interpret as the *multi-agent causal graph*. The question: **how does causal invariance** in the multiway system lead to *maximal TE* among those vantage processes $\{X_t^{(i)}\}$?

33 Statement of the Theorem

Theorem (Causal Invariance $\implies$ TE Convergence).

Let $\mathcal{W} = (\mathcal{H}, \rho, h_0)$ be a multiway rewriting system with N partial vantage observers $\{\mathcal{A}_1, \dots, \mathcal{A}_N\}$.

Suppose $\mathcal{W}$ is *strongly causally invariant*:

For any two vantage branches Π_i and Π_j (the event paths that produce local time series for $\mathcal{A}_i, \mathcal{A}_j$), there is a stage M beyond which Π_i and Π_j become isomorphic in event partial order, up to a bijection ϕ on states.

Then, if Π_i, Π_j *unify* (i.e., from time M onward, they yield local states ϕ -equivalent), the Transfer Entropy

$$T_{X^{(i)} \rightarrow X^{(j)}}(t), \quad T_{X^{(j)} \rightarrow X^{(i)}}(t)$$

converge to *maximal* or near-maximal values as $t \rightarrow \infty$.

In simpler words, if two observer vantage branches unify in a strongly causally invariant system, $\{X^{(i)}\}$ and $\{X^{(j)}\}$ become effectively “the same time series” (up to a 1-1 mapping) from some step onward, forcing TE in both directions to saturate.

34 Proof

34.1 Formalization of Branch Unification

1. Let Π_i be the sequence of states (or rewriting events) producing $\{X_t^{(i)}\}$. Because of *time inhomogeneity*, each step $t \mapsto t + 1$ for agent $\mathcal{A}_i$ might skip multiple global rewriting steps or might compress them into one.
2. Let Π_j be similarly defined for $\mathcal{A}_j$.
3. **Unification** means: $\exists M$ such that for $\forall t \geq M$, the portion of the hypergraph that $\mathcal{A}_i$ sees is isomorphic to the portion $\mathcal{A}_j$ sees, via a bijection ϕ . Equivalently, the rewriting events that produce $X_t^{(i)}$ and $X_t^{(j)}$ coincide in the causal net, up to a label mapping.

34.2 Consequence for Local States

By definition of vantage isomorphism, for each time $t \geq M$:

$$X_t^{(j)} = \phi(X_t^{(i)}),$$

where ϕ is a fixed injective mapping from the state space of $\mathcal{A}_i$ to that of $\mathcal{A}_j$. If ϕ is surjective or nearly so, then the random variables $X_t^{(i)}$ and $X_t^{(j)}$ are in one-to-one correspondence.

34.3 Transfer Entropy under 1-1 Mapping

Definition: The TE from agent i to agent j at step t can be written in standard form:

$$T_{X^{(i)} \rightarrow X^{(j)}}(t) = H(X_{t+1}^{(j)} | X_t^{(j)}) - H(X_{t+1}^{(j)} | X_t^{(j)}, X_t^{(i)}).$$

(We assume a discrete state space for simplicity.)

Since for $t \geq M$, we have $X_t^{(j)} = \phi(X_t^{(i)})$, let us see how this affects predictability:

1. If the rewriting is sufficiently deterministic or has minimal noise, knowledge of $X_t^{(i)}$ might completely fix $X_{t+1}^{(j)}$. Then $H(X_{t+1}^{(j)} | X_t^{(j)}, X_t^{(i)}) \approx 0$.
2. Meanwhile, $H(X_{t+1}^{(j)} | X_t^{(j)})$ might remain nontrivial if the local dynamics for $\mathcal{A}_j$ are not purely Markov in $\mathcal{A}_j$'s own vantage.

Hence $T_{X^{(i)} \rightarrow X^{(j)}}(t) \approx H(X_{t+1}^{(j)} | X_t^{(j)})$, possibly large. In the extreme case, if $\{X_t^{(i)}\}$ determines $\{X_{t+1}^{(j)}\}$ *exactly*, then

$$T_{X^{(i)} \rightarrow X^{(j)}}(t) = H(X_{t+1}^{(j)} | X_t^{(j)}).$$

If the latter has a large value (depending on state cardinalities, etc.), it saturates. A similar argument shows $T_{X^{(j)} \rightarrow X^{(i)}}(t)$ is also large if ϕ (or ϕ^{-1}) is well-defined.

Hence from time $t \geq M$ onward,

$$T_{X^{(i)} \rightarrow X^{(j)}}(t) > \text{some large constant},$$

and similarly for $T_{X^{(j)} \rightarrow X^{(i)}}(t)$. In many discrete setups, “maximal TE” is close to the log cardinality of the relevant next-state space.

34.4 Summation or Limit

We can define an infinite-horizon TE measure by summing or averaging over t . Because from $t \geq M$ onward TE is large, any time-average

$$\overline{T_{X^{(i)} \rightarrow X^{(j)}}} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} T_{X^{(i)} \rightarrow X^{(j)}}(t)$$

is dominated by the large post-unification region, leading to a final value near or at the maximum. This completes the argument that **causal invariance** $\implies$ *TE saturates*, provided vantage unification occurs.

35 Conclusion of the Theoretical Proof

We have shown the following:

1. **Mapping:** Each agent $\mathcal{A}_i$ is mapped to a vantage branch Π_i in the multiway system, producing a local time series $\{X_t^{(i)}\}$.
2. **Causal Invariance:** If the system’s rewriting is strongly causally invariant, Π_i and Π_j become isomorphic for large t if they unify.
3. **State Isomorphism:** This implies a 1-1 (or nearly 1-1) mapping between $\mathcal{A}_i$ and $\mathcal{A}_j$ states from some step M onward.
4. **TE Saturation:** By standard information-theoretic arguments about 1-1 correspondences, the Transfer Entropy from $\mathcal{A}_i$ to $\mathcal{A}_j$ is then “maximal or nearly maximal” for all $t \geq M$. Summing or averaging yields a global measure of high TE.

Thus, in a purely *theoretical sense*, one can indeed prove that **causal invariance** in Wolfram’s sense implies **maximal** or near-maximal **statistical dependence** among the vantage-based agent processes, if those vantage branches unify. The formal steps revolve around:

- Defining vantage branches,
- Stating the isomorphism guaranteed by causal invariance,
- Translating isomorphism into near-perfect correlation (or 1-1 function) between states,
- Using TE’s definition to show it saturates under that correlation.

Hence the statement “**causal invariance** $\implies$ **maximal TE** among unifying observers” is mathematically provable in a strictly theoretical manner, assuming the definitions and conditions in the list above are fully spelled out.

36 Bi-Directional Computational Interfacing with the Mind via Time-Inhomogeneous Markov Chains of Free-Energy Deltas

In the preceding sections, we introduced a time-inhomogeneous Markov chain for each agent $\mathcal{A}_i$, whose states $x_t^{(i)}$ represent *free-energy deltas* (ΔF) at discrete time steps t . Below, we show how these ΔF -based Markov chains can be *driven* by external *stimulation patterns* that effectively *encode or write* specific trajectories into each agent’s behavioral tendencies. Simultaneously, one can *read* from the chain states if they are observable. This yields a *bi-directional* computational interface, reminiscent of a *brain-computer interface* but grounded in an information-theoretic, time-inhomogeneous Markov model of ΔF .

36.1 Setup and Basic Definitions

Agent ΔF -Chain. Each agent $\mathcal{A}_i$ has a discrete (or continuous) state space $\mathcal{S}^{(i)}$, with elements $x_t^{(i)} \in \mathcal{S}^{(i)}$ denoting short-term changes in the agent’s free energy at time t . We posit a *time-inhomogeneous* (or nonstationary) transition kernel

$$P_t^{(i)}(x_{t+1}^{(i)} | x_t^{(i)}, \mathbf{u}_t), \quad (25)$$

where $\mathbf{u}_t \in \mathcal{U}$ is an *action* or *stimulation* chosen externally at time t . The dependence on $\mathbf{u}_t$ means we *control* or *steer* each agent’s ΔF transitions via external perturbations, possibly neural or environmental.

Multi-Agent Joint State. In a multi-agent setting, we define

$$\mathbf{x}_t = (x_t^{(1)}, \dots, x_t^{(N)}),$$

the joint configuration of all agents’ ΔF states. Then an action $\mathbf{u}_t$ updates all agents in the next step:

$$\mathbf{x}_{t+1} \sim P_t(\mathbf{x}_{t+1} | \mathbf{x}_t, \mathbf{u}_t) = \prod_{i=1}^N P_t^{(i)}(x_{t+1}^{(i)} | x_t^{(i)}, \mathbf{x}_t, \mathbf{u}_t),$$

assuming (if desired) some factorization structure. One can also allow coupling among agents’ transitions if $x_{t+1}^{(i)}$ depends on $x_t^{(j)}$ for $j \neq i$.

36.2 Reinforcement Learning Control of ΔF Trajectories

Action Space and Policy. We let $\mathbf{u}_t \in \mathcal{U}$ be the “stimulus” or “perturbation” we apply at time t . This can be:

- *Neural stimulation* signals (e.g., TMS pulses, deep-brain stimulation patterns),
- *Environmental manipulations* (e.g., sensory illusions, changes to the agent’s environment),
- *Direct brain–computer interface* signals written into the agent’s system.

A *policy* $\pi(\mathbf{u}_t | \mathbf{x}_t)$ decides which stimulation to apply in each joint state $\mathbf{x}_t$.

Reward and Objective. Define a *reward* function $r(\mathbf{x}_{t+1}, \mathbf{u}_t)$ capturing the “desirability” of a particular next state or action. For *therapeutic aims*, the reward might penalize pathological ΔF patterns (indicating extreme or unstable personality or attachment traits). For *maximal mutual correlation* among agents, the reward could incorporate cross-agent dependence measures (e.g., multi-way mutual information).

Formally, an RL problem arises: find π to maximize expected cumulative reward

$$\mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t r(\mathbf{x}_{t+1}, \mathbf{u}_t) \right],$$

where $\gamma \leq 1$ is a discount factor. By *exploring* different stimulation sequences $\{\mathbf{u}_0, \dots, \mathbf{u}_{T-1}\}$, the RL agent learns a policy that “shapes” each ΔF chain trajectory as desired.

36.3 Encoding Information in the ΔF Trajectory (the “Write” Aspect)

The transition (25) shows that choosing $\mathbf{u}_t$ influences the *distribution* of $x_{t+1}^{(i)}$. If we interpret each distinct path $(x_0^{(i)}, x_1^{(i)}, \dots)$ as a *signal* in the agent’s mind or behavior, then controlling $\{\mathbf{u}_t\}$ amounts to *encoding* a chosen trajectory:

$$\{\mathbf{u}_0, \dots, \mathbf{u}_{T-1}\} \longmapsto (x_0^{(i)}, \dots, x_T^{(i)}).$$

Hence, the chain is a “communication channel,” and $\mathbf{u}_t$ is an “input” that modifies the agent’s “output” (the ΔF states). One can design $\mathbf{u}_t$ patterns to systematically *write* into the chain—especially if we track the chain’s current state $x_t^{(i)}$ in real time.

36.4 Reading the Chain (the “Read” Aspect)

If we can *measure* or *infer* $x_t^{(i)}$ at each step, we gather knowledge of the agent’s ΔF trajectory—akin to reading from the agent’s mind/behavior:

- *Direct readout*: If the agent’s ΔF is externally measurable (e.g., from physiological signals).
- *Posterior inference*: If $x_t^{(i)}$ is hidden, we do a smoothing or filtering procedure.

This yields a *bi-directional interface*: we *write* by selecting $\mathbf{u}_t$, and we *read* by observing $x_t^{(i)}$.

36.5 Multi-Agent Coupling and “Objective Reality” via Shared Stimulations

Consider N agents, each with ΔF chain $x_t^{(i)}$. The RL engine can produce *shared* or *coordinated* stimulation $\mathbf{u}_t$, leading to updates

$$x_{t+1}^{(i)} \sim P_t^{(i)}(\cdot | x_t^{(i)}, \mathbf{u}_t), \quad i = 1, \dots, N.$$

If we aim to *maximize mutual correlation* among all $\{x_t^{(i)}\}$, we define an objective function like

$$\mathcal{J}(\pi) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \left[\sum_{i < j} I(x_t^{(i)}, x_t^{(j)}) \right],$$

where $I(\cdot, \cdot)$ is (for instance) mutual information, or we might use Transfer Entropy. By *solving* for π^* , we get a stimulation policy that *orchestrates* each chain to move in lockstep, effectively creating a “*shared reality*”. Indeed, when something “changes,” it changes for *all* agents in a correlated manner.

Therapy or AR/VR Example. 1. *Therapy*: If extremes in ΔF transitions (due to personality or attachment dysfunction) are recognized as “pathological,” the RL reward will penalize those transitions. Stimulation is found that systematically shift each agent’s chain *away* from pathological loops or spikes—facilitating a stable or “healthy” region in ΔF . 2. *AR/VR*: We choose $\mathbf{u}_t$ that push multiple agents’ ΔF states to reflect illusions or consistent storylines. Everyone “experiences” the same illusions, thus forging an *artificial* objective reality.

36.6 Subjective vs. Objective Reality: Relaxing the Constraints

If we do not want a fully “shared reality,” we allow each agent to have partially independent stimuli or environment. Then the correlation among $x_t^{(i)}$ is partial or lower. The RL objective might aim to blend “common illusions” with “agent-specific illusions.” This can systematically create *different subjective realities*.

36.7 Formal Connections to Markov Blankets

In Active Inference, an agent’s “internal states” are conditionally independent from “environment states” except via a “Markov blanket.” In a *discrete-time* setting with ΔF chains:

- Let X_t^{int} be internal states capturing the agent’s hidden ΔF “beliefs” or impetus,
- Let X_t^{ext} be external environment states,
- Let B_t be the blanket subset (e.g., active/sensory states).

The chain transitions

$$p_t(X_{t+1}^{\text{int}}, B_{t+1}, X_{t+1}^{\text{ext}} | X_t^{\text{int}}, B_t, X_t^{\text{ext}})$$

are factorized or constrained to reflect that int and ext only influence each other through B_t . In the *stimulation* setting, $\mathbf{u}_t$ also enters as a control variable that modifies B_t or X_t^{ext} or both. Then *steering the chain* is equivalently steering how the blanket mediates internal–external interactions.

Hence, from a Markov-blanket perspective, *controlling* B_t or X_t^{ext} over time modifies the agent’s ΔF transitions or internal dynamics. This merges the *Active Inference* viewpoint with a *time-inhomogeneous chain + RL* approach, enabling precise “digital pharmaceuticals” or illusions by carefully selecting $\mathbf{u}_t$.

37 Reinforcement Learning Control of Cognitive State Trajectories via Matrix Transfer Entropy (MTE)

In this section, we present a framework for controlling the evolution of an agent’s cognitive state—represented as a coarse-grained free-energy delta (ΔF) state—using reinforcement learning (RL). Our approach leverages Matrix Transfer Entropy (MTE) to quantify the influence of external stimulations on the agent’s state transitions, ensuring that the control does not overwhelm the agent’s intrinsic dynamics. We first describe the single-agent case and then extend the formulation to a multi-agent context.

37.1 Single-Agent Formulation

37.1.1 Agent Dynamics and State Representation

We assume that an agent $\mathcal{A}$ has a finite set of coarse-grained cognitive states:

$$\mathcal{S} = \{s_1, s_2, \dots, s_M\},$$

where each state $x_t \in \mathcal{S}$ corresponds to a distinct free-energy delta (ΔF) level, capturing the agent’s cognitive configuration at time t .

The evolution of the agent is modeled by a time-inhomogeneous Markov chain with an external control signal. Specifically, the transition dynamics are given by:

$$P_t(x_{t+1} \mid x_t, u_t),$$

where $u_t \in \mathcal{U}$ is the external stimulation applied at time t . The time inhomogeneity reflects that the transition probabilities may change with time (due to environmental factors or internal modulation).

37.1.2 Reinforcement Learning Problem Setup

The RL controller’s objective is to determine a policy

$$\pi(u_t \mid x_t)$$

that selects stimulations u_t based on the current state x_t so as to drive the agent’s state transitions toward a desired pattern. Rather than targeting a specific state, we target a desired *transition dynamic* P^* representing, for example, healthy or adaptive cognitive trajectories.

Reward Function We design a composite reward $r(x_{t+1}, u_t)$ that comprises three components:

- (i) **Trajectory Tracking Reward:** This term penalizes deviations from the desired transition behavior. If P_t is the empirical transition pattern observed over a window, we define

$$r_{\text{trans}} = -D(P_t \parallel P^*),$$

where $D(\cdot \parallel \cdot)$ is a divergence measure (e.g., Kullback-Leibler divergence).

- (ii) **Self-Dynamics Preservation Reward:** To prevent excessive perturbation, we penalize deviations from the agent’s natural (uncontrolled) dynamics P_{nat} :

$$r_{\text{self}} = -D(P_t \parallel P_{\text{nat}}).$$

- (iii) **MTE-Based Influence Reward:** Matrix Transfer Entropy (MTE) quantifies the influence of the stimulation u_t on the agent’s state transition. Define the embedded state and stimulation vectors:

$$\mathbf{x}_t^{(k)} = [x_t, x_{t-1}, \dots, x_{t-k+1}],$$

$$\mathbf{u}_t^{(k)} = [u_t, u_{t-1}, \dots, u_{t-k+1}].$$

Then, the MTE from the stimulation to the agent’s state is given by

$$\text{MTE}_{u \rightarrow x} = S_2(x_t \mid \mathbf{x}_t^{(k)}) - S_2(x_t \mid \mathbf{x}_t^{(k)}, \mathbf{u}_t^{(k)}),$$

where $S_2(\cdot)$ is the 2nd-order matrix entropy computed via Gram matrices with a Gaussian kernel. We set a target influence level, $\text{MTE}_{u \rightarrow x}^{\text{target}}$, and define

$$r_{\text{MTE}} = -\left| \text{MTE}_{u \rightarrow x}^{\text{current}} - \text{MTE}_{u \rightarrow x}^{\text{target}} \right|.$$

The overall reward is a weighted combination:

$$r(x_{t+1}, u_t) = \alpha r_{\text{trans}} + \beta r_{\text{self}} + \gamma r_{\text{MTE}},$$

with α , β , and γ balancing the objectives.

RL Objective The RL problem is to maximize the expected cumulative reward:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t r(x_{t+1}, u_t) \right],$$

where $\gamma \in (0, 1]$ is the discount factor.

37.1.3 Learning and Feedback

During online learning, the policy $\pi(u_t \mid x_t)$ is continuously updated using observed state transitions and computed MTE. This real-time feedback ensures that if the stimulation is too weak or too strong (as measured by the deviation from the target MTE), the corresponding reward term penalizes the action, guiding the RL agent toward an optimal balance between modifying transition dynamics and preserving the agent’s inherent behavior.

37.2 Extension to the Multi-Agent Context

Consider a network of agents $\{\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_N\}$, where each agent $\mathcal{A}_i$ has its own coarse-grained state $x_t^{(i)} \in \mathcal{S}^{(i)}$. We define the joint state:

$$\mathbf{x}_t = (x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(N)}),$$

and a joint stimulation vector:

$$\mathbf{u}_t = (u_t^{(1)}, u_t^{(2)}, \dots, u_t^{(N)}).$$

37.2.1 Joint Transition Dynamics

Each agent’s transition dynamics may depend on both its own state and on the states of other agents:

$$x_{t+1}^{(i)} \sim P_t^{(i)}(x_{t+1}^{(i)} \mid x_t^{(i)}, \mathbf{x}_t, \mathbf{u}_t).$$

This framework permits inter-agent coupling and coordinated behaviors.

37.2.2 Incorporating Multi-Agent MTE

To foster a shared or coordinated experience, we introduce an additional reward term based on the mutual influence among agents. For each pair of agents i and j , we compute the pairwise MTE:

$$\text{MTE}_{i \rightarrow j} = S_2(x_t^{(j)} \mid \mathbf{x}_t^{(j)(k)}) - S_2(x_t^{(j)} \mid \mathbf{x}_t^{(j)(k)}, \mathbf{x}_t^{(i)(k)}),$$

where $\mathbf{x}_t^{(i)(k)}$ is an embedding of the past k states of agent i .

An aggregate multi-agent MTE reward is then defined as:

$$r_{\text{MTE-multi}}(t) = \frac{2}{N(N-1)} \sum_{i < j} [\text{MTE}_{i \rightarrow j}(t) + \text{MTE}_{j \rightarrow i}(t)].$$

This term encourages the stimulation patterns and resulting transitions to exhibit strong mutual dependence, analogous to aligning content or experiences across agents.

37.2.3 Composite Multi-Agent Reward and Policy Learning

The overall reward function for the multi-agent RL system becomes:

$$r(t) = \alpha r_{\text{trans}}(t) + \beta r_{\text{self}}(t) + \gamma r_{\text{MTE-single}}(t) + \delta r_{\text{MTE-multi}}(t),$$

where:

- $r_{\text{MTE-single}}(t)$ is defined as in the single-agent case (ensuring each agent receives the appropriate external influence), and
- $r_{\text{MTE-multi}}(t)$ promotes coordinated transitions across agents.

A centralized (or coordinated decentralized) RL policy π is then learned to select joint stimulation signals:

$$\pi(\mathbf{u}_t \mid \mathbf{x}_t),$$

maximizing the expected cumulative reward:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t r(t) \right].$$

This formulation drives each agent’s transitions toward desired patterns while simultaneously aligning their experiences by maximizing mutual influence (as measured by MTE).

Conclusion

We have presented a unified framework that first addresses the single-agent scenario, where RL is employed to steer an agent’s cognitive state transitions (represented as coarse-grained ΔF states) using MTE as an influence measure. The extension to the multi-agent context adds a coordination mechanism by incorporating an additional reward term that maximizes the statistical dependence between agents. This integrated approach ensures that interventions modify transition dynamics in a controlled manner—guiding the agent toward desired behaviors without overwhelming its intrinsic self-regulatory processes—while fostering aligned experiences across multiple agents.

We have presented a rigorous, formal scheme for using time-inhomogeneous Markov chains (states = free-energy deltas) as the basis of a *bi-directional computational interface* with the mind. The essential steps are:

- (i) **Define Markov chains** whose states track each agent’s ΔF changes over time.
- (ii) **Incorporate an action/stimulation space $\mathbf{u}_t$** that influences transitions, forming an RL control problem.
- (iii) **Devise an objective** that e.g. aims for certain therapeutic or shared illusions, or for maximizing multi-agent correlation to replicate “objective reality.”
- (iv) **Implement reading/writing:** read the chain states (or infer them), apply $\mathbf{u}_t$ to “write” new directions in the chain.

- (v) **(Optional) Markov Blanket framing:** interpret external vs. internal states parted by a blanket, consistent with Active Inference’s separation of environment and agent.

This framework can unify:

- *Therapy* (steering away from pathology via RL feedback),
- *Multi-agent illusions* or *virtual realities* (shared ΔF transitions),
- *Subjective vs. objective* manipulations, as one can tune how strongly agent states remain correlated.

Thus, we see how *time-inhomogeneous ΔF -based Markov chains* become a novel “communication channel” to *write into* and *read from* each agent’s mental or behavioral processes, bridging computational RL stimulation with the notion of “free-energy-based mind states.”

38 Discussion and Future Directions

The framework of Computational Psychodynamics presented in this paper opens numerous avenues for future research and practical applications. Here we outline several key initiatives that will extend and validate this framework across multiple domains.

38.1 Forthcoming Technical Implementation Paper

A comprehensive follow-up paper is in development that will provide a detailed treatment of the computational implementation of Computational Psychodynamics. The GitHub repository linked will contain the open-source development of this implementation. This technical paper will address:

1. Practical Implementation:

- Detailed code implementations for various computational models
- Guidelines for integration into existing systems
- Framework validation through simulation studies

2. Methodological Guidelines:

- Best practices for implementing the framework in research contexts
- Validation methodologies and metrics
- Replication protocols for experimental studies

3. System Integration:

- Methods for incorporating the framework into existing psychological assessment tools
- Integration patterns for clinical applications
- Approaches for scaling implementations across different domains

This technical paper will serve as a practical guide for researchers and practitioners looking to implement and extend the Computational Psychodynamics framework in their own work. Initial guides will be added to the appendix for people to use before the follow-up paper.

38.2 Major Development Projects

38.2.1 Generative Story Worlds

The Generative Story Worlds project represents a significant application of Computational Psychodynamics in narrative generation and interactive storytelling. This initiative focuses on:

1. Character Development:
 - Implementation of personality models derived from Computational Psychodynamics
 - Dynamic character behavior generation based on personality states
 - Interactive character evolution through narrative progression
2. Narrative Architecture:
 - Mathematical formalization of narrative dynamics
 - Integration of state-based personality transitions
 - Development of coherent, personality-driven plot progressions
3. System Integration:
 - Implementation of state-of-the-art narrative modeling frameworks
 - Development of novel interactive storytelling mechanisms
 - Creation of robust evaluation metrics for narrative coherence

This project aims to demonstrate the practical application of Computational Psychodynamics in creating rich, dynamic narrative environments with psychologically realistic characters. Generative Story Worlds can be found in the implementation in the Github Repository for Computational Psychodynamics.

38.2.2 Project Universum: Brain-Computer Interface Development

Project Universum represents an extension of Computational Psychodynamics into the domain of neural interfaces and cognitive enhancement. This initiative encompasses:

1. Neurofeedback Development:
 - Integration of reinforcement learning mechanisms
 - Development of targeted transcranial magnetic stimulation protocols that can encode information to write to minds
 - Creation of adaptive neurofeedback systems
2. Neural State Modeling:
 - Extension of Computational Psychodynamics to perceptual state modeling
 - Integration of machine learning with neuroimaging data
 - Development of real-time neural state prediction systems
3. Technical Integration:
 - Implementation of EEG-based monitoring systems
 - Development of fMRI-compatible modeling approaches
 - Creation of multimodal neural recording frameworks

This project aims to bridge the gap between theoretical frameworks and practical neural interface applications, potentially revolutionizing our approach to brain-computer interfaces.

38.3 Future Research Directions

The development of these projects and papers will likely generate additional research questions and opportunities, including:

1. Theoretical Extensions:

- Integration with emerging cognitive science frameworks
- Development of new mathematical models for psychological processes
- Extension to group dynamics and social systems

2. Validation Studies:

- Large-scale empirical validation of the framework
- Cross-cultural application studies
- Longitudinal effectiveness research

3. Clinical Applications:

- Development of therapeutic applications
- Integration with existing clinical frameworks
- Creation of diagnostic tools

The future development of Computational Psychodynamics through these initiatives promises to expand our understanding of human cognition and behavior while providing practical tools for various applications. These projects represent the beginning of a broader research program that could fundamentally transform our approach to psychological modeling and intervention.

39 Conclusion

This paper has presented Computational Psychodynamics, a novel mathematical framework that bridges crucial gaps in psychological theory and practice. By integrating principles from Active Inference, ecological psychology, and cybernetic theories with sophisticated mathematical tools, we have developed a comprehensive approach to understanding and modeling human behavior, cognition, and personality.

The framework’s foundation rests on several key innovations. First, the development of observation-based psychometrics provides a more ecological and objective approach to personality assessment, moving beyond the limitations of traditional self-report measures. Second, our adaptation of the Free Energy Principle and Active Inference to personality modeling offers a principled mathematical basis for understanding behavioral dynamics, grounded in the fundamental principle of least action.

The hierarchical schema we have presented, based on linearly separable binary classifications, provides a structured approach to understanding behavioral states and transitions. This schema’s connection to time-inhomogeneous Markov chains and hidden Markov models enables sophisticated modeling of both observable behaviors and latent psychological states. The framework’s ability to capture transfer entropy between these behavioral chains offers novel insights into attachment patterns, social influences, and personality dynamics.

Perhaps most significantly, Computational Psychodynamics provides a unified theoretical framework that connects seemingly disparate aspects of psychological theory, computational neuroscience, and physics. From ecological approaches to personality assessment to cybernetic theories of behavior, from attachment theory to the Big Five personality traits, our framework demonstrates how these various perspectives can be integrated into a coherent mathematical structure. This integration not only advances our theoretical understanding but also enables practical applications in areas ranging from artificial intelligence to clinical psychology.

The framework’s versatility is particularly evident in its potential applications. Whether modeling character behaviors in narrative generation, developing brain-computer interfaces, or analyzing social network dynamics, Computational Psychodynamics provides the mathematical tools and theoretical foundation necessary for sophisticated psychological modeling and prediction.

As we look to the future, this framework opens new avenues for research and application in psychology and related fields. By providing a mathematical basis for understanding human behavior and cognition, Computational Psychodynamics not only advances our theoretical understanding but also enables practical applications that could transform how we approach psychological assessment, intervention, and modeling.

The true potential of this framework lies in its ability to bridge the gap between theory and practice, between mathematical modeling and psychological understanding. As we continue to develop and refine these approaches, we anticipate that Computational Psychodynamics will contribute to a more nuanced, precise, and practical understanding of human psychology, ultimately leading to more effective interventions and applications across multiple domains.

Acknowledgments

I thank Emily Hasson for creating and introducing the diagram of perplexity and (self versus other) analogous to the entropy axis formulation with self versus other in Binary Cognitive Mechanics but an important contribution for theoretical clarity on what entropy is describing in Binary Cognitive Mechanics. I thank Emily Hasson and Dr. Stephen Culpepper for introducing me to the use of Hidden Markov Chains for modeling behavioral states. I thank Dr. Daniel Friedman for insights into the Active Inference framework and for looking over earlier versions of this manuscript. I also thank Dr. Steven Read for teaching me a wide breadth of motivational theory. I would like to thank Binyamin Tsadik and the ByCog Group for introducing me the the Binary Cognitive Mechanics System. I thank Lamar McNeil for introducing and teaching me the Objective Personality System and Jungian Theory. I also thank Yash Addaganti for helpful discussions related to the Jungian Theory.